THE CIRCULAR LAW FOR INDEPENDENT ROW SLICES DOWN TO LOGARITHMIC DEGREE

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ABSTRACT. Let M_n be an $n \times n$ random 0–1 matrix with independent rows, each uniform over vectors having exactly d ones. We prove that, for an absolute constant $C_* > 0$, the empirical spectral distribution of

$$\frac{M_n}{\sqrt{d(1-d/n)}}$$

converges in probability to the circular law whenever $C_* \log n \le d$ and $d = o(n)$. This lowers the known $\log^{2+\varepsilon} n$ threshold to sufficiently large logarithmic degree throughout the sparse range.

The main obstacle is the lack of column regularity, which removes the aggregate-profile identities available for random regular digraphs. In the lower polylogarithmic range, we replace those identities by an averaged multinomial row-profile estimate and combine it with a rectangular right-kernel analysis to control all but $O(n/d^2)$ of the smallest singular values. A regularized row-replacement argument then compares logarithmic determinants with those of an i.i.d. Bernoulli matrix. Above every fixed superlogarithmic cutoff, the same comparison is combined instead with a uniform polynomial smallest-singular-value bound. Girko hermitization and the sparse i.i.d. circular law complete the proof.

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Date: Revised July 2026.

2020 *Mathematics Subject Classification.* 60B20, 15B52, 05C80, 60F05.

Key words and phrases. circular law, sparse random matrices, row-regular matrices, intermediate singular values, random normals, small-ball probability.

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1. Introduction

1.1. **The model and the main result.** For integers $1 \le d \le n$, let

$$\Omega_{n,d} := \left\{ x \in \{0,1\}^n : \sum_{j=1}^{n} x_j = d \right\}.$$

We consider an $n \times n$ random matrix M_n whose rows are independent and each is uniformly distributed on $\Omega_{n,d}$. Thus every row has exactly d nonzero entries, while the column sums fluctuate. We call this the *independent row-slice model*.

For a square matrix A, write

$$\mu_A := \frac{1}{n} \sum_{j=1}^{n} \delta_{\lambda_j(A)}$$

for its empirical spectral distribution, with algebraic multiplicity. The circular law μ_{circ} is the uniform probability measure on the unit disk in $\mathbb{C}$.

Put $p = d/n$. Each entry of M_n has variance $p(1-p)$, so the natural bulk scale is

$$\sqrt{np(1-p)} = \sqrt{d(1-d/n)}.$$

The exact row sums also give

$$M_n \mathbf{1} = d\mathbf{1}.$$

Thus d is a deterministic eigenvalue, which becomes a deterministic outlier at

$$\sqrt{\frac{d}{1-d/n}}$$

under the bulk normalization. This single forced eigenvalue contributes only $1/n$ to the empirical spectral distribution, so it does not affect the limiting law.

The circular law for the row-slice model was established by Li, Litvak, and Yu [LLY26a] under the condition $d \geq \log^{2+\varepsilon} n$, $d = o(n)$. Their smallest-singular-value theorem already reaches the logarithmic invertibility threshold $d \geq C \log n$. The gap between these two ranges is caused by the logarithmic contribution of intermediate singular values. The purpose of this paper is to close that gap throughout the sparse range above a sufficiently large logarithmic constant.

Theorem 1.1 (Main theorem)**.** *There exists an absolute constant $C_* > 0$ with the following property. Let $d = d(n)$ be any integer sequence satisfying*

$$C_* \log n \leq d, \qquad d = o(n).$$

If M_n is the independent row-slice matrix described above, then

$$\mu_{M_n/\sqrt{d(1-d/n)}} \xrightarrow{\mathbb{P}} \mu_{\mathrm{circ}}.$$

The constant C_* is chosen large enough for the support-diameter estimate in the max-weight lemma, the imported smallest-singular-value and restricted-norm estimates, and the rectangular witness and net arguments in Section 4; see remarks 3.4 and 4.5. The proof does not establish the result for every positive constant multiplying $\log n$.

The proof uses two complementary regimes. For every fixed $\sigma \in (0, 1)$, the new structural and intermediate-singular-value argument is uniform for

$$C_* \log n \leq d \leq (\log n)^{1+\sigma}. \tag{1.1}$$

For every fixed $\varepsilon > 0$, a separate superlogarithmic bridge treats

$$(\log n)^{1+\varepsilon} \leq d, \qquad d = o(n). \tag{1.2}$$

The two regimes use the same regularized row-replacement comparison but different controls on the bottom singular values:

Degree regime	Prefix codimension	Bottom-tail input
$C_* \log n \leq d \leq (\log n)^{1+\sigma}$	$r = \left\lceil n/\sqrt{d} \right\rceil$	new intermediate-singular-value bound
$(\log n)^{1+\varepsilon} \leq d$, $d = o(n)$	$r = \lceil n/d^{a_\varepsilon} \rceil$	uniform polynomial smallest-singular-value bound

Here

$$a_\varepsilon = \frac{2+\varepsilon}{2(1+\varepsilon)}.$$

In both rows, the comparison transfers the relevant prefix log volume to an i.i.d. Bernoulli matrix; interlacing and hermitization then recover the square-matrix spectral law.

For example, take $\sigma = \varepsilon = 1/2$ and split an arbitrary sequence at $(\log n)^{3/2}$. The lower-band theorem applies below the cutoff and the superlogarithmic bridge applies above it. This two-regime argument yields

theorem 1.1. More generally, any common fixed exponent may be used. For degrees above $\log^{2+\eta} n$, with any fixed $\eta \in (0,1)$, the circular-law theorem of Li–Litvak–Yu [LLY26a, Theorem 1.3] also applies directly, providing an independent overlap with the bridge rather than a third logically necessary piece of the proof.

1.2. **Context.** Girko introduced the hermitization and logarithmic-potential approach to the circular law in [Gir85]; modern accounts include [BC12, TV10]. For matrices with independent entries, the classical sequence of results includes Bai's theorem [Bai97] and the optimal finite-variance circular law of Tao and Vu [TV10]. In the sparse i.i.d. Bernoulli model, the optimal divergence condition $np \to \infty$ is known from [RT19, SSS25].

Circular-law questions with fixed linear constraints have a separate history. Nguyen and Vu proved the circular law for matrices whose rows are independent random $\{-1,1\}$ vectors with a prescribed row sum [NV13]. For directed matrices with both row and column sums fixed, the sparse circular law was established in [Coo19, LLTTY21]. Related invertibility and singularity results for combinatorial and sparse matrices include [Tra20, FKS22, LLY26a]. We abbreviate Litvak–Lytova–Tikhomirov–Tomczak-Jaegermann–Youssef as LLTTY.

The present independent row-slice model has exact row sums and independent rows, but no exact column regularity. Li, Litvak, and Yu proved its circular law when $d \geq \log^{2+\varepsilon} n$, $d = o(n)$, and obtained smallest-singular-value estimates down to $d \geq C \log n$ [LLY26a]. The remaining gap is therefore not an invertibility threshold; it is the logarithmic contribution of a block of intermediate singular values. In the doubly regular model, exact column regularity supplies deterministic identities for aggregate row profiles. Those identities disappear here because the column sums fluctuate, and replacing them is the main new task.

1.3. **Proof architecture and new ingredients.** The proof separates imported mechanisms from arguments specific to the independent row-slice model:

Imported interfaces	New arguments proved here
LLTTY discretization and gradual-vector decomposition	Averaging the one-row multinomial profile before tensorization
LLTTY fixed-subspace Gaussian-normal estimates	Direct rectangular exclusion of the shifted steep and almost-constant classes
Li–Litvak–Yu smallest-singular-value and restricted-norm estimates	Exact-slice double counting and the $O(n/d^2)$ intermediate-singular-value bound
Sparse i.i.d. Bernoulli circular law and hermitization interface	Arbitrary-codimension regularized row replacement and the two-regime synthesis

Fix $\sigma \in (0,1)$ and work first in the lower band (1.1). Set

$$m = \left\lfloor n/d^3 \right\rfloor - 1, \qquad \gamma = \frac{1}{288}, \qquad \rho = \exp[-C(n/m)^\gamma], \qquad \delta_{\mathrm{cl}} = (m/n)^{\gamma/2}.$$

Thus $(n/m)^\gamma = d^{1/96+o(1)}$ and $\delta_{\mathrm{cl}} = d^{-1/192+o(1)}$. The lower-band proof has six steps.

1. Averaged row profiles. For a canonical approximation y, let B be the dyadic order of the truncated LLTTY weight attached to a uniformly chosen coordinate. A with-replacement row has a multinomial profile Q, and the LLTTY one-row small-ball estimate remains valid conditional on Q. Averaging before tensorization gives

$$\mathbb{E}_Q \,\mathrm{SB}(Q) \le C e^{-cD(B)} \prod_q \widetilde{w}_q^{-p_q} + e^{-cd^{2/3}}, \qquad D(B) = \mathbb{E}\left|B - B'\right|,$$

where B' is an independent copy. The key input is

$$\mathbb{E} 2^{-\max_{a \le d} B_a} \le 2^{-\mathbb{E}B} \exp\left(-c\mathbb{E}\left|B - B'\right|\right)$$

under the support-diameter condition proved in Section 3. Conditioning the sampled coordinates to be distinct costs only $e^{O(d^2)} = e^{o(n)}$.

2. Structural control of rectangular kernels. The averaged estimate is inserted into the LLTTY decomposition

$$S_L = \bigcup_{u=4}^{v} K_u(L) \cup P_v(L).$$

In the K_u cases, either the truncated weights already supply a factor d^{-cn} or two constant-mass weight populations are separated by cd^κ. In the clustered P_v case, the separation is comparable to v, where

$$v \asymp \frac{d^{1/96}}{\log d}, \qquad \delta_{\mathrm{cl}} v \gg \log d.$$

The resulting energy factor pays for the remaining profile entropy. A separate rectangular right-kernel argument excludes the codimension-dependent steep and almost-constant classes and identifies the remaining vectors with the exact normalized gradual class. This yields the row-slice structural theorem for codimensions m and $m+1$.

3. Random normals and intermediate singular values. The Gaussian-normal argument of [LLTTY21, Section 4] is model-independent once the exposed row span has the structural property. Outside $O(n\delta_{\mathrm{cl}})$ exceptional coordinates, a Gaussian normal has no ρ-disk containing more than m coordinates. Exact-slice double counting then gives one-row concentration

$$O(m/n) + e^{-cd} = O(d^{-3}).$$

Leave-one-row-out distances and the negative second-moment identity give the following bound.

Theorem 1.2 (Intermediate singular values)**.** *Fix $\sigma \in (0,1)$ and assume (1.1). For every fixed $z \in \mathbb{C}$, there are constants $C_{\mathrm{isv}}(z), C'_{\mathrm{isv}}(z) > 0$ and integers $q_0 = q_0(n,z)$ with*

$$q_0 \le C'_{\mathrm{isv}}(z)\frac{n}{d^2}$$

such that

$$\mathbb{P}\left\{ s_{n-q_0}(M_n - z\sqrt{d}\, I_n) \ge e^{-C_{\mathrm{isv}}(z) d^{1/96}} \right\} = 1 - o(1).$$

The quantitative feature needed later is the deletion count $q_0 = O(n/d^2)$, not merely $q_0 = o(n)$. At the lower-band prefix codimension $r \asymp n/\sqrt{d}$, only the bottom q_0 singular values pay the full smallest-singular-value cost, while the remaining omitted singular values pay the intermediate scale. After division by n, the two negative-log contributions are

$$O\Big(\frac{\log n \, \log\log n}{d^2}\Big) = o(1), \qquad O\Big(d^{-1/2} d^{1/96}\Big) = O(d^{-47/96}) = o(1).$$

This is the point at which the $O(n/d^2)$ bound closes the circular-law argument at logarithmic degree.

4. Regularized row replacement. For a prefix with $k = n - r$ rows, a Schur-complement hybrid reduces each row replacement to a regularized distance associated with a positive contraction of trace at least $r+1$. Bernoulli convex concentration followed by conditioning on the exact row sum yields

$$\frac{1}{n}\mathbb{E}\,|\Psi_\eta(H_{\mathrm{slice}}) - \Psi_\eta(H_{\mathrm{iid}})| \le C\sqrt{\frac{\log(ed)}{Q}} + C\sqrt{d}\log\frac{U}{\eta}\, e^{-cQ},$$
$$Q = p(1-p)(r+1).$$

For the lower band, $r = \left\lceil n/\sqrt{d} \right\rceil$ gives $Q \asymp \sqrt{d}$.

5. Lower-band hermitization. Only $q_0 = O(n/d^2)$ singular values pay the full smallest-singular-value cost. The intermediate bound controls the rest of the omitted block. Weak singular-measure convergence, full logarithmic-potential convergence, and Hilbert–Schmidt control give the corresponding Bernoulli bottom-tail estimate. Interlacing then converts the prefix comparison into a square log-determinant comparison, and the replacement principle closes the lower band.

6. The superlogarithmic bridge. For fixed $\varepsilon > 0$, the refined Li–Litvak–Yu smallest-singular-value threshold is uniformly polynomial when $d \ge (\log n)^{1+\varepsilon}$. With

$$a_\varepsilon = \frac{2+\varepsilon}{2(1+\varepsilon)}, \qquad r = \lceil n/d^{a_\varepsilon} \rceil,$$

one has

$$Q \asymp d^{1-a_\varepsilon} \to \infty, \qquad d^{-a_\varepsilon}\log n \le (\log n)^{-\varepsilon/2} = o(1).$$

Thus all r omitted singular values may pay the polynomial lower bound, so the same replacement theorem covers the full superlogarithmic range. Its overlap with the lower-band theorem proves theorem 1.1.

The logical dependency of the lower-band argument may be summarized as

$$\begin{gathered}\text{row-local small ball} \Longrightarrow \text{averaged profiles},\\ \text{averaged profiles} \Longrightarrow \text{rectangular structure},\\ \text{rectangular structure} \Longrightarrow \text{Gaussian normals},\\ \text{Gaussian normals} \Longrightarrow \text{intermediate singular values},\\ \text{intermediate singular values} \Longrightarrow \text{regularized replacement},\\ \text{regularized replacement} \Longrightarrow \text{determinants} \Longrightarrow \text{circular law}.\end{gathered}$$

Appendices A and B give the source map and probability bookkeeping for each arrow.

1.4. **Organization.** Section 2 fixes notation, the lower-band regime, and the imported deterministic and probabilistic ingredients. Section 3 proves the averaged profile estimate. Section 4 proves the rectangular right-kernel compatibility theorem for the exact LLTTY classes. Section 5 closes the K_u and P_v branches and derives the row-slice structural theorem. Section 6 treats Gaussian normals and exact-slice anti-concentration. Section 7 proves theorem 1.2. Section 8 proves the generalized regularized row-replacement estimate. Section 9 proves the lower-band circular law, establishes the superlogarithmic bridge, and assembles theorem 1.1. Section 10 discusses the remaining sublogarithmic obstruction. The appendices give a lemma-by-lemma extraction of the LLTTY deterministic consequences and a dependency ledger for all imported results.

2. Preliminaries and imported ingredients

2.1. **Notation.** All logarithms are natural unless a base is displayed. Constants denoted by c, C, C' are positive and may change from line to line. A constant carrying a subscript may depend on that subscript but not on n.

Fix a parameter $\sigma \in (0,1)$. Throughout the structural argument in Sections 3 to 7 and the lower-band hermitization argument, the integer degree satisfies

$$C_* \log n \le d \le (\log n)^{1+\sigma}. \tag{2.1}$$

All constants in those arguments are uniform in d on this interval; the threshold from which the estimates hold may depend on σ. The superlogarithmic bridge in Section 9.7 is stated separately and does not assume (2.1). Rounding is suppressed only inside asymptotic relations such as $m = n/d^3 + O(1)$.

Constant convention. Let $a_3^{\mathrm{src}}, c_K, c_P, c_{4.10} > 0$ denote fixed admissible constants from the cited LLTTY statements. Choose once and for all

$$0 < a_3 \le a_3^{\mathrm{src}}$$

sufficiently small for the entropy and row-load inequalities in Sections 4.7 and 4.8. This is permitted because the source construction takes a_3 to be a sufficiently small absolute constant; throughout the manuscript the LLTTY classes and the scale $n_3 = \lfloor a_3 n \rfloor$ are defined using this single final choice. Next choose the absolute normal parameter $a \le c_{4.10}$, and then enlarge the absolute constant C_* so that the support-diameter estimate, the imported smallest-singular-value and restricted-norm estimates, and the rectangular witness and net estimates all hold. After these absolute choices, fix $\sigma \in (0,1)$ and a shift bound $Z_0 < \infty$. Only the threshold

$$n \ge n_0(\sigma, Z_0)$$

may depend on these two parameters. No later argument further decreases a_3 or enlarges a source constant.

For a vector $x \in \mathbb{C}^n$, let

$$x_1^* \ge x_2^* \ge \cdots \ge x_n^*$$

be the nonincreasing rearrangement of $(|x_j|)_{j \le n}$. For a matrix H with at least as many columns as rows, its singular values are denoted

$$s_1(H) \ge \cdots \ge s_{\min\{\#\text{rows},\#\text{columns}\}}(H) \ge 0.$$

For a $k \times n$ matrix H, define its row volume by

$$\mathrm{Vol}(H) := \sqrt{\det(HH^*)} = \prod_{j=1}^{k} s_j(H).$$

We use

$$\log^+ t := \max\{\log t, 0\}, \qquad \log^- t := \max\{-\log t, 0\}.$$

For probability measures μ, ν on $\mathbb{C}$, write

$$d_{\mathrm{BL}}(\mu,\nu) := \sup\left\{ \left| \int f\, d\mu - \int f\, d\nu \right| : \|f\|_\infty \le 1,\ \mathrm{Lip}(f) \le 1 \right\}.$$

For random probability measures, $\mu_n \Rightarrow \mu$ in probability means $d_{\mathrm{BL}}(\mu_n, \mu) \to 0$ in probability. The same convention is used for differences of two random empirical spectral measures.

Our Hermitian inner product is

$$\langle u, v \rangle = \sum_{j=1}^{n} u_j \overline{v_j},$$

linear in the first variable. Matrix rows act bilinearly:

$$r \cdot x = \sum_{j=1}^{n} r_j x_j.$$

Thus $r \cdot x = \langle x, \overline{r} \rangle$. This convention is used throughout the right-kernel, Gaussian-normal, and leave-one-row-out sections.

A row slice is a random vector X uniformly distributed on $\Omega_{n,d}$. We write $p = d/n$. A Bernoulli comparator row is denoted by

$$\boldsymbol{\xi} = (\xi_1, \dots, \xi_n), \qquad \xi_j \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

We also write

$$\mathbb{C}^\times := \mathbb{C} \setminus \{0\}, \qquad \mathbb{C}^\times \mathcal{X} := \{\lambda x : \lambda \in \mathbb{C}^\times,\ x \in \mathcal{X}\}$$

for the nonzero scalar cone generated by a set $\mathcal{X} \subset \mathbb{C}^n$.

2.2. **Discretization and the LLTTY ℓ-decomposition.** We recall only the portions of the discretization from [LLTTY19, Sections 5–7] that enter our proof.

Definition 2.1 (k-vector)**.** For an integer $k \geq 1$, a vector $y \in \mathbb{C}^n$ is a k-vector if $ky_j \in \mathbb{Z} + i\mathbb{Z}$ for every j. The canonical k-approximation of $x \in \mathbb{C}^n$ is

$$y_j = \frac{1}{k}(\lfloor k\Re x_j \rfloor + i \lfloor k\Im x_j \rfloor).$$

Given a k-vector y, its coordinates are partitioned into level sets and then into regular and spread ℓ-parts by the source construction. At order $j \geq 0$, let I_j be the coordinates not assigned at an earlier order and let $\Lambda_j = \{y_i : i \in I_j\}$. For $\lambda \in \Lambda_j$, the level set $L(j, \lambda)$ is the entire remaining λ-level when that level has cardinality less than 2^{j+1}, and otherwise is its leftmost subset of cardinality 2^j. Thus every nonempty $L(j, \lambda)$ has cardinality between 2^{j-1} and 2^{j+1}.

The spread values at order j are chosen as follows. If

$$\operatorname{diam}(\Lambda_j) < d/k,$$

then the spread family is empty and every level set of order j is assigned to the regular part. Otherwise the canonical greedy procedure of [LLTTY19, Section 5.2] selects a maximal pairwise d/k-separated family $\Lambda_j^S \subset \Lambda_j$, starting from a separated pair. In particular, Λ_j^S then contains at least two values. The spread and regular parts of order j are

$$\mathcal{L}_j^S = \bigcup_{\lambda \in \Lambda_j^S} L(j, \lambda), \qquad \mathcal{L}_j^R = \bigcup_{\lambda \in \Lambda_j \setminus \Lambda_j^S} L(j, \lambda).$$

Consequently, a singleton family of represented values is always regular, every nonempty spread part has height at least two, and the distinct values represented in a spread part are pairwise d/k-separated. Every value represented in a regular part is either in a set of diameter less than d/k (when $\mathcal{L}_j^S = \varnothing$) or lies at distance less than d/k from the greedy spread family.

Empty parts are omitted and the nonempty spread and regular parts are re-enumerated as

$$\mathcal{L}^{(1)}, \dots, \mathcal{L}^{(r)}, \qquad r \leq 3 \log n.$$

The height $h_q = h(\mathcal{L}^{(q)})$ is the number of level sets comprising $\mathcal{L}^{(q)}$. If $\mathcal{L}^{(q)}$ has order j, then

$$2^{j-1}h_q \le s_q := |\mathcal{L}^{(q)}| \le 2^{j+1}h_q. \tag{2.2}$$

We use the source's lexicographic tie-breaking and ordering of the nonempty parts, which make the decomposition canonical. After permuting coordinates, the types, orders, sizes, and heights of the parts agree up to relabeling. Only these invariants are used below.

For $p_q = s_q/n$ and a profile count $Q_q \in \{0, \dots, d\}$, define the actual weights

$$w_q(Q) = \begin{cases} h_q Q_q/d, & \mathcal{L}^{(q)} \text{ regular}, \\ h_q\sqrt{Q_q}, & \mathcal{L}^{(q)} \text{ spread}, \end{cases} \tag{2.3}$$

and the truncated weights

$$\widetilde{w}_q = \begin{cases} h_q p_q, & s_q \ge nd^{-1/3},\ \mathcal{L}^{(q)} \text{ regular}, \\ h_q/d, & s_q < nd^{-1/3},\ \mathcal{L}^{(q)} \text{ regular}, \\ h_q\sqrt{dp_q}, & s_q \ge nd^{-1/3},\ \mathcal{L}^{(q)} \text{ spread}, \\ h_q, & s_q < nd^{-1/3},\ \mathcal{L}^{(q)} \text{ spread}. \end{cases} \tag{2.4}$$

In particular,

$$\frac{h_q}{d} \le \widetilde{w}_q \le n\sqrt{d}. \tag{2.5}$$

The dyadic w-set of order b is the union of those $\mathcal{L}^{(q)}$ for which

$$2^b \le \widetilde{w}_q < 2^{b+1}.$$

The conditional one-row estimator is

$$\mathrm{SB}(Q) := \min\left(1, \min_{q:Q_q>0} w_q(Q)^{-1}\right), \tag{2.6}$$

and its truncated analogue is

$$\mathrm{SB}(\widetilde{Q}) := \min\left(1, \min_{q:Q_q>0} \widetilde{w}_q^{-1}\right). \tag{2.7}$$

2.3. **Author-derived deterministic consequences of the LLTTY decomposition.** The following result is extracted from [LLTTY19, Lemmas 5.1, 5.4, 5.5, 7.2, 7.7–7.9 and Theorem 5.6]. Its K_u and clustered-P_v conclusions include intermediate estimates from the proofs of the cited source lemmas, not merely verbatim source statements. Appendix A gives a lemma-by-lemma derivation, with exact source-version, page, lemma, and equation locators. The source works in the degree range $d \le \exp(c\sqrt{\log n})$ for an absolute $c > 0$. Under (2.1), every fixed polylogarithmic upper bound is eventually smaller than this quantity, so the global source restriction is automatic.

In the lower-band argument, set

$$m := \left\lfloor n/d^3 \right\rfloor - 1$$

and call a codimension admissible if

$$L \in \{m, m+1\} = \left\{ \left\lfloor n/d^3 \right\rfloor - 1, \left\lfloor n/d^3 \right\rfloor \right\}.$$

For all sufficiently large n, one has $m \geq 1$, and every admissible codimension satisfies the source restriction

$$1 \leq L \leq \frac{n}{d^3}.$$

For each admissible codimension L, let S_L denote the exact normalized gradual class of [LLTTY19, Section 4], formed with the codimension-dependent steep classes and the final constant a_3 fixed above. This is the source class used in Sections 5 to 7; Section 4.2 reproduces its defining inequalities in full. Thus no statement below depends on a definition deferred to a later section.

For $x \in \mathbb{C}^n$, a scale $u > 0$, and $\rho > 0$, a set $J \subset [n]$ is a ρ-cluster of x at scale u if there is $\lambda \in \mathbb{C}$ such that

$$|x_j - \lambda| \leq \rho u \qquad (j \in J).$$

When $u = x^*_{n3}$, we simply say that x has a ρ-cluster of cardinality $|J|$.

For an integer $k \geq 1$, let

$$\mathcal{A}_k(L) := \{y : y \text{ is the canonical } k\text{-approximation of some } x \in S_L\}.$$

Fix the lexicographic order on $\mathbb{C}$, let $y^\sharp$ denote the coordinates of y rearranged in nonincreasing lexicographic order, and, for $y, \widetilde{y} \in \mathcal{A}_k(L)$, set

$$y \sim_k \widetilde{y} \quad \Longleftrightarrow \quad y^\sharp = \widetilde{y}^\sharp.$$

Thus an equivalence class consists precisely of normalized canonical approximants having the same coordinate multiset; this relation is not asserted on the unrestricted, infinite set of all k-vectors. For $\mathcal{X} \subset S_L$, write

$$\mathcal{A}_k(\mathcal{X}) := \left\{ y : \begin{array}{l} y \text{ is the canonical } k\text{-approximation} \\ \text{of some } x \in \mathcal{X} \end{array} \right\} \subset \mathcal{A}_k(L).$$

and use the restriction of $\sim_k$ to this set.

Proposition 2.2 (Author-derived deterministic consequences of the LLTTY decomposition)**.** *There are absolute constants $a_3, c_K, c_P, c_0, c_1 > 0$ such that, for every admissible codimension L and the exact normalized gradual class S_L, there are families $K_u(L), P_u(L) \subset S_L$ with the following properties.*

(i) *For every integer $v \geq 5$,*

$$S_L = \bigcup_{u=4}^{v} K_u(L) \cup P_v(L).$$

(ii) *Suppose additionally that $d \leq n^{1/3}$ and*

$$1 \leq k \leq \frac{\sqrt{n}}{d^{3/2}}.$$

Then $\mathcal{A}_k(L)$ has at most e^n equivalence classes under $\sim_k$. The same bound holds for the nonempty classes obtained by restricting $\sim_k$ to any

$\mathcal{A}_k(\mathcal{X}) \subset \mathcal{A}_k(L)$. *If* $\mathcal{C}$ *is such a class with* ℓ*-part sizes* s_q *and heights* h_q*, then*

$$|\mathcal{C}| \le \frac{n!}{\prod_q s_q!} \prod_q h_q^{s_q}. \tag{2.8}$$

(iii) *If* $I(b) = \{q : 2^b \le \widetilde{w}_q < 2^{b+1}\}$ *and* $t_b = \sum_{q \in I(b)} s_q$*, then*

$$\prod_{q \in I(b)} s_q! \ge t_b! e^{-Ct_b}. \tag{2.9}$$

Moreover, for $J(b) \subset I(b)$ *consisting of parts of size* $< d^{-\kappa}n$*, where*

$$\kappa = \frac{c_K a_3}{16} < \frac{1}{3},$$

one has

$$\left| \bigcup_{q \in J(b)} \mathcal{L}^{(q)} \right| \le C d^{-\kappa} n. \tag{2.10}$$

(iv) *If* y *is a* d^u*-approximation of a vector in* $K_u(L)$*, then the total mass of spread* ℓ*-parts is at least* $c_K n_3$*, where* $n_3 = \lfloor a_3 n \rfloor$*. With*

$$I = \{q : s_q \ge d^{-\kappa} n\},$$

one of the two alternatives holds:

$$\sum_{q \in I} s_q \ge n - \frac{c_K n_3}{4}, \tag{K1}$$

$$\sum_{q \in I^c} s_q \ge \frac{c_K n_3}{4}. \tag{K2}$$

In alternative (K1)*,*

$$\prod_q h_q^{s_q} \prod_q \widetilde{w}_q^{-s_q} \le C^n d^{-c_1 n}. \tag{2.11}$$

In alternative (K2)*, there are integers* $b_1 \le b_2$ *such that the coordinate distribution of the dyadic weight order has mass at least* c_0 *below* b_1 *and at least* c_0 *above* b_2*, and*

$$b_2 - b_1 \ge c_1 d^{\kappa}. \tag{2.12}$$

(v) *Fix* $v \ge 5$ *and* $\rho \le d^{-v}$*. Assume that*

$$0 < \delta \le 36 c_P a_3 \tag{2.13}$$

and

$$c_P a_3 (v-4) \ge 2\log_2 d + 2\log_2\left(\frac{72\sqrt{d}}{\delta}\right) + 2 - \log_2(c_P a_3). \tag{2.14}$$

If $x \in P_v(L)$ *has a* ρ*-cluster of at least* δn *coordinates and* y *is its* d^v*-approximation, then the dyadic weight-order distribution has mass at least* $\delta/36$ *below some* b_1*, mass at least* c_0 *above some* b_2*, and*

$$b_2 - b_1 \ge \frac{c_P a_3}{2}(v-4). \tag{2.15}$$

The source status of the five parts is summarized below. Page numbers refer to arXiv:1801.05575v4.

Part	Status	Exact locator
(i)	verbatim decomposition after notation change	p. 38, Theorem 5.6
(ii)	direct restatement and reindexing	pp. 34–35, Lemma 5.1 and (14)–(16); p. 37, Lemmas 5.4–5.5
(iii)	stated factorial bound plus proof-level cutoff consequence	p. 55, Lemma 7.2; p. 61, Case 2 in the proof of Lemma 7.7
(iv)	author-derived packaging of the two proof branches	pp. 60–61, proof of Lemma 7.7, equations (32)–(33), (38), Cases 1–2
(v)	author-derived combination of two named lemmas	p. 62, Lemma 7.8; pp. 62–63, Lemma 7.9 and its proof

Remark 2.3. Appendix A shows in particular that (2.11) is the three-family calculation in the first case of [LLTTY19, Lemma 7.7], while the separated weight populations in (K2) and part (v) are intermediate conclusions used in the proofs of [LLTTY19, Lemmas 7.7 and 7.9]. The factorial estimate (2.9) and the small-bin estimate (2.10) use exactly the source's truncated weights and w-sets; no change of binning is made.

2.4. **Row-slice inputs from Li–Litvak–Yu.** We use the following consequences of [LLY26a].

Proposition 2.4 (Smallest singular values)**.** *There are absolute constants $C, C_1, C_2 > 0$ such that, whenever $C \log n \le d \le n/2$ and $|\zeta| \le \sqrt{d} \log\log d$, both the independent row-slice matrix M_n and the i.i.d. Bernoulli(d/n) matrix B_n satisfy*

$$\mathbb{P}\left\{ s_n(M_n - \zeta I_n) \le n^{-\beta_n} \right\} + \mathbb{P}\left\{ s_n(B_n - \zeta I_n) \le n^{-\widetilde{\beta}_n} \right\} \le \frac{C_1}{\sqrt{d}},$$

for exponents

$$2 < \beta_n, \widetilde{\beta}_n \le C_2 \log\log n.$$

This combines [LLY26a, Theorem 1.7, Remark 1.9, Theorem 3.1 and Remark 3.2].

Corollary 2.5 (Uniform polynomial scale above a superlogarithmic cutoff)**.** *Fix $\varepsilon > 0$. There exists a constant $B_\varepsilon < \infty$ such that, whenever*

$$(\log n)^{1+\varepsilon} \le d \le n/2, \qquad |\zeta| \le \sqrt{d} \log\log d,$$

one has

$$\mathbb{P}\left\{ s_n(M_n - \zeta I_n) \le n^{-B_\varepsilon} \right\} + \mathbb{P}\left\{ s_n(B_n - \zeta I_n) \le n^{-B_\varepsilon} \right\} \le \frac{C}{\sqrt{d}}. \tag{2.16}$$

Proof. We make the extraction from the source explicit. Write

$$\Gamma(n,d) := \frac{2\log(6d)}{\log(d/(C_2 \log n))}.$$

In the sparse branch of [LLY26a, Remark 1.9], the admissible threshold, denoted there by $\alpha(n,d)$, satisfies

$$\alpha(n,d) \geq n^{-2\Gamma(n,d)}.$$

If $d \geq (\log n)^{1+\varepsilon}$, then for all sufficiently large n,

$$\log \frac{d}{C_2 \log n} = \log d - \log\log n - \log C_2 \geq \frac{\varepsilon}{2(1+\varepsilon)} \log d.$$

Also $\log(6d) \leq 2\log d$ for large d. Consequently,

$$\Gamma(n,d) \leq \frac{8(1+\varepsilon)}{\varepsilon}$$

uniformly throughout the sparse branch. Hence the source threshold there is at least n^{-B_ε} for any fixed

$$B_\varepsilon \geq \frac{16(1+\varepsilon)}{\varepsilon}.$$

In the complementary dense branch, the same remark gives

$$\alpha(n,d) \geq \frac{1}{2nd^{3/2}} \geq n^{-3}$$

for all sufficiently large n, since $d \leq n/2$. Increasing B_ε if necessary therefore gives a single polynomial threshold throughout $d \leq n/2$. Remark 3.2 of the same paper permits exactly the same threshold for the Bernoulli comparator. The probability estimate is the one in proposition 2.4. □

Proposition 2.6 (Restricted operator norm in the range used below)**.** *Let H be an $m \times n$ matrix with independent uniform d-subset rows. Assume*

$$\frac{n}{2} \leq m \leq n, \qquad d \leq \frac{n}{2}.$$

For every fixed $A > 0$ there is $C_A < \infty$ such that, whenever $d \geq C_A \log n$,

$$\|H - \mathbb{E}H\| \leq C_A\sqrt{d}$$

with probability at least $1 - n^{-A}$.

Proof. Apply [LLY26a, Theorem 4.1] with its parameters $\gamma = A + 4$ and $\beta = A + 1$. The source failure probability is

$$nm^{-\gamma} + 2n^{-\beta}.$$

Since $m \geq n/2$, this is at most n^{-A} for all sufficiently large n. Write

$$N = \max\{9d, 2\gamma \log m\}.$$

Because $m \leq n$ and $d \geq C_A \log n$, enlarging C_A if necessary gives $N = 9d$. The source estimate

$$\|H - \mathbb{E}H\| \leq \frac{2d}{\sqrt{N}} + C_1\beta\sqrt{N}$$

therefore yields $\|H - \mathbb{E}H\| \leq C_A\sqrt{d}$. This is the only range of the rectangular theorem used in the manuscript. □

The right-kernel exclusions needed later are not imported: their complete rectangular proof is given in Section 4.

2.5. **Concentration and tensorization.** We use the Lévy concentration function

$$\mathcal{Q}(\xi, t) := \sup_{\lambda \in \mathbb{C}} \mathbb{P}\left\{|\xi - \lambda| \le t\right\}.$$

The following standard tensorization statement is the form used in [LLTTY19, Lemma 2.7].

Lemma 2.7 (Tensorization)**.** *Let $\xi_1, \dots, \xi_K$ be independent complex random variables. If*

$$\mathcal{Q}(\xi_i, \tau t) \le A\tau^2 \qquad \text{for all } \tau \ge 1 \text{ and all } i,$$

then, for an absolute C and every $\tau \ge 1$,

$$\mathbb{P}\left\{\sum_{i=1}^{K} |\xi_i|^2 \le C\tau^2 t^2 K\right\} \le (CA\tau^2)^K.$$

The statement remains valid after deterministic shifts of the ξ_i.

We also use the negative second-moment identity. If H is a full-row-rank $k \times n$ matrix with rows R_i, then

$$\sum_{j=1}^{k} s_j(H)^{-2} = \sum_{i=1}^{k} \operatorname{dist}\big(R_i, \operatorname{span}(R_\ell : \ell \ne i)\big)^{-2}. \tag{2.17}$$

The inequality needed later remains valid after zero singular values are removed, or after restricting to the surviving full-row-rank submatrix.

3. An averaged row-profile small-ball estimate

This section contains the principal model-specific input. The regular-digraph proof in [LLTTY19] conditions on a global profile matrix and then uses exact column regularity to cancel profile entropy. In the independent row-slice model, global profiles fluctuate. We instead average the one-row estimator over the multinomial profile before tensorizing over the independent rows.

3.1. **The with-replacement representation.** Let y be a k-vector with nonempty ℓ-parts $\mathcal{L}^{(1)}, \dots, \mathcal{L}^{(r)}$. Put

$$s_q = |\mathcal{L}^{(q)}|, \qquad p_q = s_q/n.$$

Let $J_1, \dots, J_d$ be i.i.d. uniform random variables on $[n]$ and define

$$\widehat{X} := \sum_{a=1}^{d} e_{J_a}.$$

Its profile relative to the ℓ-decomposition is

$$Q_q := \#\left\{a \le d : J_a \in \mathcal{L}^{(q)}\right\}, \qquad Q \sim \text{Multinomial}(d; p_1, \dots, p_r).$$

Conditional on Q, the samples belonging to distinct ℓ-parts are independent, and within $\mathcal{L}^{(q)}$ they are i.i.d. uniform on that part. If $\psi_{q,a}$ denotes the corresponding level value of y, then

$$\widehat{X} \cdot y \stackrel{d}{=} \sum_{q=1}^{r} \sum_{a=1}^{Q_q} \psi_{q,a} \qquad \text{conditional on } Q. \tag{3.1}$$

Lemma 3.1 (Conditional row-local small ball)**.** *For every profile Q and every $\tau \geq 1$,*

$$\mathcal{Q}\left(\widehat{X} \cdot y, \frac{\tau\sqrt{d}}{k} \,\middle|\, Q \right) \leq C\tau^2 \,\mathrm{SB}(Q).$$

Proof. This is the row-local content of [LLTTY19, Lemma 6.4]. We indicate why no global regularity enters. The blocks in (3.1) are independent, so convolution reduces the estimate to any visited part q.

The level sets inside one ℓ-part have cardinalities within a factor four, hence

$$\max_z \mathbb{P}\{\psi_{q,a} = z\} \leq \frac{4}{h_q + 3}.$$

For a regular part, the distinct represented values form a $1/k$-separated subset of the complex plane. The discrete planar concentration estimate [LLTTY19, Proposition 2.5] gives

$$\mathcal{Q}\left(\sum_{a=1}^{Q_q} \psi_{q,a}, \frac{\tau\sqrt{d}}{k} \right) \leq C \max\left\{ \frac{d\tau^2}{h_q Q_q}, \frac{1}{h_q} \right\} \leq \frac{C\tau^2}{w_q(Q)}.$$

For a spread part, the distinct represented values are d/k-separated and, by the source-compatible construction in Section 2, one has $h_q \geq 2$. In the Miroshnikov branch $h_q > 32$, we may first assume

$$1 \leq \tau^2 \leq w_q(Q) = h_q\sqrt{Q_q},$$

since otherwise $C\tau^2/w_q(Q) \geq 1$, after taking $C \geq 1$, and the desired probability bound is trivial. Under this reduction, $Q_q \leq d$, the atom bound above, and the d/k-separation give, for all sufficiently large d,

$$\max_{a \leq Q_q} \mathcal{Q}\left(\psi_{q,a}, \frac{\tau\sqrt{2d}}{k} \right) \leq \frac{8 + 64\tau^2/d}{h_q} \leq \frac{1}{2}.$$

This verifies the atom hypothesis in the Miroshnikov estimate. Applying the two-dimensional Esseen estimate [LLTTY19, Proposition 2.2] when $2 \leq h_q \leq 32$, and the Miroshnikov estimate [LLTTY19, Proposition 2.3] when $h_q > 32$, yields

$$\mathcal{Q}\left(\sum_{a=1}^{Q_q} \psi_{q,a}, \frac{\tau\sqrt{d}}{k} \right) \leq \frac{C\tau^2}{h_q\sqrt{Q_q}} = \frac{C\tau^2}{w_q(Q)}.$$

The lower bound $h_q \geq 2$ is essential here: a one-level block is deterministic conditional on Q_q and therefore belongs to the regular part. For such a block

$w_q(Q) = Q_q/d \le 1$, so the preceding regular estimate and the truncation in (2.6) give the required trivial bound.

Taking the minimum over all visited parts, and truncating the bound by one, proves the lemma. The proof uses only the counts Q_q, the conditional independence in (3.1), and the deterministic geometry of the ℓ-decomposition. The identities $\sum_i Q_{iq} = ds_q$ and exact column regularity do not occur. □

3.2. **Comparison with truncated weights.** Define the large-part set

$$\mathscr{L} := \left\{ q : p_q \ge d^{-1/3} \right\}.$$

For $q \in \mathscr{L}$, the marginal Q_q is binomial with mean $dp_q \ge d^{2/3}$. Let

$$\mathcal{G} := \bigcap_{q \in \mathscr{L}} \left\{ Q_q \ge \tfrac{1}{2} dp_q \right\}.$$

Since $|\mathscr{L}| \le d^{1/3}$, Chernoff's inequality gives

$$\mathbb{P}(\mathcal{G}^c) \le d^{1/3} e^{-cd^{2/3}} \le e^{-c'd^{2/3}}. \tag{3.2}$$

On $\mathcal{G}$, the definitions (2.3)–(2.4) imply

$$w_q(Q) \ge \frac{1}{2} \widetilde{w}_q$$

for every visited large regular part, the same inequality with $1/\sqrt{2}$ for every visited large spread part, and $w_q(Q) \ge \widetilde{w}_q$ for every visited small part. Hence

$$\mathbb{E}_Q \operatorname{SB}(Q) \le 2\mathbb{E}_Q \operatorname{SB}(\widetilde{Q}) + e^{-cd^{2/3}}. \tag{3.3}$$

3.3. **The max-weight inequality.** Let B be a real random variable and let $B_1, \dots, B_d$ be i.i.d. copies. The next lemma is the mechanism that converts occupancy of weight bins into both a geometric mean and an entropy-dispersion penalty.

Lemma 3.2 (Max-weight lemma)**.** *There are absolute constants $\varepsilon_0, c > 0$ such that the following holds for every integer $d \ge 2$. If*

$$R := \operatorname{diam}(\operatorname{supp} B) \le \varepsilon_0 d,$$

then

$$\mathbb{E} 2^{-\max_{a \le d} B_a} \le 2^{-\mathbb{E}B} \exp\bigl(-c\mathbb{E}|B - B'|\bigr), \tag{3.4}$$

where B' is an independent copy of B.

Proof. Put $\mu = \mathbb{E}B$ and $D = \mathbb{E}|B - B'|$. Let $t = \lfloor d/2 \rfloor$ and pair the first $2t$ samples. Set

$$Z_j = \max(B_{2j-1}, B_{2j}), \qquad 1 \le j \le t.$$

Since $\max_{a \le d} B_a \ge t^{-1} \sum_{j=1}^{t} Z_j$, with $\theta = (\log 2)/t$ we have

$$\mathbb{E} 2^{-\max B_a} \le \bigl(\mathbb{E} e^{-\theta Z_1}\bigr)^t. \tag{3.5}$$

The identity

$$\max(B, B') = \frac{B + B' + |B - B'|}{2}$$

gives

$$\mathbb{E}Z_1 = \mu + \frac{D}{2}. \tag{3.6}$$

By Efron–Stein and the fact that the maximum is 1-Lipschitz in each coordinate,

$$\operatorname{Var}(Z_1) \le \mathbb{E}(B - B')^2 \le RD.$$

The bounded-variable Bernstein moment estimate yields, whenever $\theta R \le 1/2$,

$$\log \mathbb{E}e^{-\theta(Z_1 - \mathbb{E}Z_1)} \le C\theta^2 RD.$$

Indeed, $|Z_1 - \mathbb{E}Z_1| \le R$; expanding the exponential and summing the terms of order at least two gives

$$\mathbb{E}e^{-\theta(Z_1 - \mathbb{E}Z_1)} \le 1 + C\theta^2 \operatorname{Var}(Z_1),$$

and $\operatorname{Var}(Z_1) \le RD$. Combining this with (3.5) and (3.6), and using $t \ge d/3$, gives

$$\log \mathbb{E}2^{-\max B_a} \le -(\log 2)\mu - \frac{\log 2}{2} D + C\frac{R}{d} D.$$

Choosing ε_0 sufficiently small absorbs the last term and proves (3.4). □

3.4. **Averaging the profile.** Define B by selecting an ℓ-part according to coordinate mass:

$$\mathbb{P}\{B = \lfloor \log_2 \widetilde{w}_q \rfloor\} = p_q, \qquad D(B) := \mathbb{E}|B - B'|. \tag{3.7}$$

Also set

$$W_y := \prod_{q=1}^{r} \widetilde{w}_q^{-p_q}. \tag{3.8}$$

If J_a falls in part q_a, then

$$\mathrm{SB}(\widetilde{Q}) \le \frac{1}{\max_{a \le d} \widetilde{w}_{q_a}} \le 2^{-\max_{a \le d} B_a}.$$

Moreover,

$$2^{-\mathbb{E}B} \le 2 \prod_{q} \widetilde{w}_q^{-p_q} = 2W_y.$$

Thus lemma 3.2 and (3.3) prove the main proposition.

Proposition 3.3 (Averaged row-profile estimate)**.** *There are absolute constants $\varepsilon_0, c, C > 0$ such that, whenever*

$$\operatorname{diam}(\operatorname{supp} B) \le \varepsilon_0 d,$$

one has

$$\mathbb{E}_Q \mathrm{SB}(Q) \le Ce^{-cD(B)} W_y + e^{-cd^{2/3}}. \tag{3.9}$$

Consequently, for every $\tau \ge 1$,

$$\mathcal{Q}\left(\widehat{X} \cdot y, \frac{\tau\sqrt{d}}{k}\right) \le C\tau^2 \left(Ce^{-cD(B)} W_y + e^{-cd^{2/3}}\right). \tag{3.10}$$

The same estimate holds after adding a deterministic scalar.

Proof. Equation (3.9) follows from (3.3) and lemma 3.2. Averaging lemma 3.1 over Q gives (3.10). □

3.5. **Tensorization and conditioning to the slice.** Let $\widehat{X}_1, \dots, \widehat{X}_K$ be independent with-replacement rows and let $\widehat{M}_K$ be the resulting $K \times n$ matrix. By lemma 2.7 and proposition 3.3, for deterministic $V \in \mathbb{C}^K$,

$$\mathbb{P}\left\{ \left\| \widehat{M}_K y + V \right\|_2 \le \frac{\tau\sqrt{dK}}{k} \right\} \le \left[C\tau^2 \left(Ce^{-cD(B)} W_y + e^{-cd^{2/3}} \right) \right]^K. \quad (3.11)$$

Taking

$$\tau = C\sqrt{\frac{n}{K}} \log d$$

yields

$$\mathbb{P}\left\{ \left\| \widehat{M}_K y + V \right\|_2 \le C \log d \frac{\sqrt{dn}}{k} \right\} \le \left[C \frac{n}{K} \log^2 d \times \left(Ce^{-cD(B)} W_y + e^{-cd^{2/3}} \right) \right]^K. \quad (3.12)$$

An ordered with-replacement row has distinct samples with probability $(n)_d/n^d$, where $(n)_d = n(n-1)\cdots(n-d+1)$. Conditional on distinctness, its indicator is a uniform d-subset row. Therefore, for every event $\mathcal{E}$ depending on K independent rows,

$$\mathbb{P}_{\text{slice}}(\mathcal{E}) \le \left(\frac{n^d}{(n)_d} \right)^K \mathbb{P}_{\text{wr}}(\mathcal{E}). \quad (3.13)$$

Under (2.1), $K \le n$ and $d = o(\sqrt{n})$, so

$$K \log \frac{n^d}{(n)_d} = K \sum_{j=0}^{d-1} -\log\left(1 - \frac{j}{n}\right) = O\left(\frac{Kd^2}{n}\right) = O(d^2). \quad (3.14)$$

Hence the total conditioning cost is $e^{O(d^2)} = e^{o(n)}$, uniformly in the lower band.

Remark 3.4 (Why the logarithmic lower constant must be large). By (2.5),

$$\operatorname{diam}(\operatorname{supp} B) \le \log_2 n + \frac{3}{2} \log_2 d + 2.$$

Under (2.1),

$$\frac{\operatorname{diam}(\operatorname{supp} B)}{d} \le \frac{1}{C_* \log 2} + o(1).$$

Thus the hypothesis of lemma 3.2 holds uniformly throughout (2.1) once C_* is a sufficiently large absolute constant. This is the new profile-theoretic reason for requiring a large constant multiplying $\log n$. The final C_* must also dominate the constants in the imported smallest-singular-value and

restricted-norm inputs and in the rectangular right-kernel argument; see remark 4.5.

4. Rectangular right-kernel exclusions

This section connects rectangular right kernels in the independent-row model to the exact gradual class of [LLTTY19]. The result is uniform for the two codimensions in the leave-one-row-out argument; no square theorem is invoked.

The proof has three stages. We first exclude the shifted very-steep classes using independent-row witness events and center-invariant equal-block anti-concentration. We then treat the almost-constant remainder on a uniform row-load event. Finally, the deterministic LLTTY alternatives identify every remaining nonzero kernel vector with either the permitted very-steep class or a scalar multiple of the normalized gradual class.

4.1. **Complex convention and row action.** We use the Hermitian inner product

$$\langle u, v\rangle := \sum_{j=1}^{n} u_j \overline{v_j},$$

linear in the first variable. For matrix multiplication we use the bilinear row action

$$r \cdot x := \sum_{j=1}^{n} r_j x_j.$$

Thus $r \cdot x = \langle x, \overline{r}\rangle$. If

$$r_i = X_i - \zeta e_i, \qquad i \in I,$$

and

$$E_I := \operatorname{span}\left\{\overline{r_i} : i \in I\right\},$$

then

$$E_I^{\perp} = \ker A_{\zeta,I}, \qquad (A_{\zeta,I}x)_i = r_i \cdot x = \sum_{j=1}^{n} X_{ij}x_j - \zeta x_i. \tag{4.1}$$

No rank assumption is involved in (4.1).

4.2. **The codimension-dependent LLTTY classes.** Assume the lower-band condition (2.1) and set

$$m = \left\lfloor \frac{n}{d^3} \right\rfloor - 1, \qquad L \in \{m, m+1\}, \qquad K = n - L. \tag{4.2}$$

Let $I \subset [n]$ be deterministic with $|I| = K$, and assume throughout that $|\zeta| \le Z_0\sqrt{d}$. Write

$$P_I : \mathbb{C}^n \longrightarrow \mathbb{C}^I, \qquad P_I x = x_I, \qquad A_{\zeta,I} := M_I - \zeta P_I,$$

where M_I is the restriction of M_n to the rows indexed by I. We record the classes from [LLTTY19, Section 4], displaying their dependence on L. To avoid collision with the Bernoulli density $p = d/n$, set

$$\varepsilon_{\mathrm{st}} = \sqrt{\frac{\log d}{d}}, \qquad p_{\mathrm{st}} = \left\lfloor \frac{1}{5\varepsilon_{\mathrm{st}}} \right\rfloor = \left\lfloor \frac{1}{5}\sqrt{\frac{d}{\log d}} \right\rfloor. \tag{4.3}$$

With the fixed absolute constant $a_3 > 0$ from the constant convention, put

$$n_1 = \left\lceil \frac{n}{d^{3/2}} \right\rceil, \qquad n_2 = \left\lfloor \frac{n}{d^{2/3}} \right\rfloor, \qquad n_3 = \lfloor a_3 n \rfloor. \tag{4.4}$$

Let r be determined by

$$p_{\mathrm{st}}^{r} < n_1 \le p_{\mathrm{st}}^{r+1}, \tag{4.5}$$

and let $r_0 = r_0(L)$ be the least nonnegative integer for which

$$p_{\mathrm{st}}^{r_0} \ge \frac{20L}{d}. \tag{4.6}$$

For all sufficiently large n, uniformly for $L = m, m+1$, one has $r_0 < r$. Indeed,

$$p_{\mathrm{st}}^{r-1} \ge \frac{n_1}{p_{\mathrm{st}}^2} \ge \frac{25n\log d}{d^{5/2}}, \qquad \frac{20L}{d} \le \frac{20n}{d^4}.$$

Let

$$\mathcal{K} := \{(c, \dots, c) : c \in \mathbb{C}\}$$

be the one-dimensional space of constant vectors.

The permitted very-steep source class is

$$T_3(L) := \left\{ x : \exists i \le p_{\mathrm{st}}^{r_0},\ x_i^* > \left(\frac{n}{i}\right)^3 x^*_{p_{\mathrm{st}}^{r_0}} \right\}. \tag{4.7}$$

For $r_0 \le i \le r-1$, recursively define

$$T_{0,i}(L) := \left\{ x : x \notin T_3(L) \cup \bigcup_{j=r_0}^{i-1} T_{0,j}(L),\ x^*_{p_{\mathrm{st}}^{i}} > 4d\, x^*_{p_{\mathrm{st}}^{i+1}} \right\}. \tag{4.8}$$

At the terminal scale set

$$T_{0,r}(L) := \left\{ x : x \notin T_3(L) \cup \bigcup_{j=r_0}^{r-1} T_{0,j}(L),\ x^*_{\lceil n_1/p_{\mathrm{st}} \rceil} > 4d\, x^*_{n_1} \right\}, \tag{4.9}$$

and put

$$T_0(L) = \bigcup_{i=r_0}^{r} T_{0,i}(L). \tag{4.10}$$

The remaining source steep classes are

$$T_1(L) := \left\{ x \notin T_0(L) \cup T_3(L) : x_{n_1}^* > d^{3/2} x_{n_2}^* \right\} \tag{4.11}$$

and

$$T_2(L) := \left\{ x \notin T_0(L) \cup T_1(L) \cup T_3(L) : x_{n_2}^* > d^{3/2} x_{n_3}^* \right\}. \tag{4.12}$$

We abbreviate

$$T(L) = \bigcup_{i=0}^{3} T_i(L). \tag{4.13}$$

For $0 \le i \le 3$, define the shifted class

$$T_i^{\mathcal{K}}(L) := \left\{ v = x + y : x \in T_i(L),\ y \in \mathcal{K},\ |y_1| \le x^*_{n_1}/10 \right\}. \tag{4.14}$$

As in the source, the classes $T_0^{\mathcal{K}}$, $T_1^{\mathcal{K}}$, and $T_2^{\mathcal{K}}$ are excluded, whereas $T_3^{\mathcal{K}}$ is part of the permitted very-steep alternative.

For $\vartheta > 0$, define

$$\mathcal{B}(\vartheta) := \left\{ x : \exists \lambda \in \mathbb{C},\ \#\left\{ j : |x_j - \lambda| \le \vartheta x^*_{n_3} \right\} > n - n_3 \right\}. \tag{4.15}$$

Set

$$\vartheta_0 = \frac{10}{d^3}. \tag{4.16}$$

The normalized gradual class used in the LLTTY decomposition is literally the following.

Definition 4.1 (LLTTY gradual class)**.**

$$S_L := \left\{ x \notin T(L) \cup \mathcal{B}(\vartheta_0) : x^*_{n_3} = 1 \right\}. \tag{4.17}$$

Its dependence on L is through $r_0(L)$ and the recursive steep classes.

Let $c_{4.10} > 0$ be the absolute constant from [LLTTY21, Lemma 4.10]. Fix a sufficiently small absolute constant $a \in (0, c_{4.10}]$ for the random-normal theorem, and put

$$q_L = \lfloor aL \rfloor, \tag{4.18}$$

and define

$$V_L := \left\{ 0 \ne x : \exists i \le q_L,\ x_i^* > 0.9 \left(\frac{n}{i}\right)^3 x^*_{q_L} \right\}. \tag{4.19}$$

4.3. **The shifted T_3 class is permitted.** The deterministic inclusion from [LLTTY19, Section 4.1, equation (8)] gives

$$T_3^{\mathcal{K}}(L) \subset \left\{ v : \exists i \le p_{\mathrm{st}}^{r_0},\ v_i^* > 0.9 \left(\frac{n}{i}\right)^3 v^*_{p_{\mathrm{st}}^{r_0}} \right\}. \tag{4.20}$$

For completeness, if $v = x + c\mathbf{1}$ and $x \in T_3(L)$, then

$$|c| \le x^*_{n_1}/10 \le x^*_{p_{\mathrm{st}}^{r_0}}/10.$$

If i witnesses (4.7), order-statistic stability gives

$$v_i^* \ge x_i^* - |c|, \qquad v^*_{p_{\mathrm{st}}^{r_0}} \le x^*_{p_{\mathrm{st}}^{r_0}} + |c|.$$

Since $(n/i)^3 > 10$ for all relevant indices and all large n, the constant 0.9 in (4.20) follows.

The minimality of r_0 and (4.3) yield

$$p_{\mathrm{st}}^{r_0} < \frac{20 p_{\mathrm{st}} L}{d} \le \frac{4L}{\sqrt{d \log d}}. \tag{4.21}$$

After increasing C_*, the last quantity is at most $aL/2$. Since $L \to \infty$,

$$p_{\mathrm{st}}^{r_0} \le q_L. \tag{4.22}$$

Monotonicity of order statistics in (4.20) therefore implies

$$T_3^{\mathcal{K}}(L) \subset V_L. \tag{4.23}$$

All estimates are uniform for $L = m, m+1$.

4.4. **Unique-coordinate witnesses for $T_0^{\mathcal{K}}$.** For $r_0 \le i < r$, put

$$s_i = p_{\mathrm{st}}^i, \qquad t_i = p_{\mathrm{st}}^{i+1}. \tag{4.24}$$

At the terminal scale put

$$s_r = \left\lceil \frac{n_1}{p_{\mathrm{st}}} \right\rceil, \qquad t_r = n_1. \tag{4.25}$$

The choice $t_r = n_1$, rather than $p_{\mathrm{st}} \lceil n_1/p_{\mathrm{st}} \rceil$, is important: it matches the defining jump in (4.9) and gives direct control of the constant shift.

For disjoint sets $J^\ell, J^r \subset [n]$ satisfying $|J^\ell| = s_i$ and $|J^\ell \cup J^r| = t_i$, let

$$R = I \setminus (J^\ell \cup J^r)$$

and define

$$Z_{J^\ell, J^r} := \#\left\{ u \in R : |\operatorname{supp} X_u \cap J^\ell| = 1,\ \operatorname{supp} X_u \cap J^r = \varnothing \right\}. \tag{4.26}$$

Because the rows are independent,

$$Z_{J^\ell, J^r} \sim \mathrm{Bin}(|R|, \pi_i), \qquad \pi_i = s_i \frac{\binom{n-t_i}{d-1}}{\binom{n}{d}}.$$

Equivalently,

$$\mathbb{E} Z_{J^\ell, J^r} = |R| \frac{d s_i}{n} \prod_{h=0}^{d-2} \frac{n - t_i - h}{n - 1 - h}. \tag{4.27}$$

Since $t_i \le n_1 \le 2n/d^{3/2}$,

$$\prod_{h=0}^{d-2} \frac{n - t_i - h}{n - 1 - h} \ge 1 - \frac{(d-1)(t_i - 1)}{n - d + 1} \ge 0.9$$

for all large n. Also $|R| \ge n - L - n_1 \ge 0.9n$. Thus

$$\mathbb{E} Z_{J^\ell, J^r} \ge 0.8 d s_i, \tag{4.28}$$

and Chernoff's inequality gives

$$\mathbb{P}\left\{ Z_{J^\ell, J^r} < \tfrac{1}{2} d s_i \right\} \le e^{-c d s_i}. \tag{4.29}$$

At one scale the number of pairs is at most

$$\binom{n}{t_i} \binom{t_i}{s_i} \le \left(\frac{en}{t_i} \right)^{t_i} 2^{t_i}.$$

Here $t_i \le p_{\mathrm{st}} s_i$, including $i = r$, and

$$s_i \ge p_{\mathrm{st}}^{r_0} \ge \frac{20L}{d} \ge \frac{10n}{d^4}$$

for all large n. Therefore

$$\log\left[\binom{n}{t_i}\binom{t_i}{s_i}\right] \le C p_{\mathrm{st}} s_i \log d = o(d s_i).$$

A union bound over the $O(\log n)$ scales gives an event $\mathcal{E}_0(L)$ such that

$$\mathbb{P}(\mathcal{E}_0(L)^c) \le e^{-cn/d^3}. \tag{4.30}$$

On $\mathcal{E}_0(L)$, let $v = x + c\mathbf{1} \in T_{0,i}^{\mathcal{K}}(L)$. Order the coordinates of x by modulus and take the first s_i coordinates as J^ℓ and the next $t_i - s_i$ coordinates as J^r. Then

$$x^*_{s_i} > 4d\, x^*_{t_i}, \qquad |c| \le x^*_{n_1}/10 \le x^*_{t_i}/10. \tag{4.31}$$

For every witness row u, the row index and all selected coordinates other than the unique $j(u) \in J^\ell$ lie beyond the first t_i coordinates. Using the exact row sum,

$$|(A_{\zeta,I} v)_u| \ge x^*_{s_i} - (d - 1 + |\zeta|)x^*_{t_i} - (d + |\zeta|)|c| \ge x^*_{s_i} - 1.2d\, x^*_{t_i} \ge 0.7 x^*_{s_i} > 0$$

for all large d. Hence

$$\ker A_{\zeta,I} \cap T_0^{\mathcal{K}}(L) = \varnothing \qquad \text{on } \mathcal{E}_0(L). \tag{4.32}$$

4.5. **Uniform equal-block witness events.** Fix a sufficiently small absolute constant $c_w > 0$. For an integer $s \le n/(100d)$, let $\mathcal{W}(s)$ be the event that for every pair of disjoint sets $J^\ell, J^r \subset [n]$ with $|J^\ell| = |J^r| = s$, at least $c_w ds$ rows $i \in I$ satisfy

$$|\operatorname{supp} X_i \cap (J^\ell \cup J^r)| = 1. \tag{4.33}$$

For a fixed pair, the witness probability equals

$$2s\frac{\binom{n-2s}{d-1}}{\binom{n}{d}} \ge c\frac{ds}{n},$$

so the mean number of witnesses is at least cds. Chernoff's inequality and

$$\#\left\{(J^\ell, J^r)\right\} \le \binom{n}{2s}\binom{2s}{s} \le \exp\left(Cs \log\frac{en}{s}\right)$$

give

$$\mathbb{P}(\mathcal{W}(s)^c) \le \exp\left[-cds + Cs\log\frac{en}{s}\right]. \tag{4.34}$$

We use

$$s = n_1 \qquad \text{and} \qquad s = m_0 := \left\lfloor \frac{n}{200d} \right\rfloor. \tag{4.35}$$

At these scales, respectively,

$$\mathbb{P}(\mathcal{W}(n_1)^c) \le e^{-cn/\sqrt{d}}, \qquad \mathbb{P}(\mathcal{W}(m_0)^c) \le e^{-cn}. \tag{4.36}$$

These uniform events are kept separate from the fixed-vector small-ball probabilities below; in particular, their failures are not multiplied by net cardinalities.

4.6. **A center-invariant equal-block lemma.** The next statement is the independent-row version of the fixed-vector mechanism in [LLTTY19, Section 4.3].

Lemma 4.2 (Equal-block small ball)**.** *There is an absolute constant $C_{\mathrm{eb}} \geq 1$ such that the following holds. Let $1 \leq m_1 \leq m_2 \leq n - m_1$, and suppose that a deterministic vector $u \in \mathbb{C}^n$ satisfies*

$$u^*_{m_1} > \frac{2}{3} + u^*_j \qquad \text{for every } j > m_2. \tag{4.37}$$

Let $w \in \mathcal{K}$. Put $m_* = \min\{m_0, m_1\}$. The following estimates hold uniformly in w:

(i) *If $m_1 \leq m_0$, then*

$$\mathbb{P}\left\{\|A_{\zeta,I}(u+w)\|_2 \leq c\sqrt{dm_1},\ \mathcal{W}(m_1)\right\} \leq e^{-cdm_1}. \tag{4.38}$$

(ii) *If $m_1 \geq C_{\mathrm{eb}} m_0$, then*

$$\mathbb{P}\left\{\|A_{\zeta,I}(u+w)\|_2 \leq c\sqrt{dm_0},\ \mathcal{W}(m_0)\right\} \leq \exp\left[-cdm_0 \log\left(\frac{dm_1}{n}\right)\right]. \tag{4.39}$$

Proof. For part (i), set $q_0 = 1$ and $m_* = m_1$. For part (ii), put

$$q_0 = \left\lfloor \frac{m_1}{m_0} \right\rfloor, \qquad m_* = m_0.$$

In the second case $q_0 m_* \geq m_1/2$. Partition $q_0 m_*$ of the largest coordinates into disjoint sets $J^\ell_1, \dots, J^\ell_{q_0}$, and the same number of coordinates beyond m_2 into disjoint sets $J^r_1, \dots, J^r_{q_0}$, all of size m_*. Set $J_q = J^\ell_q \cup J^r_q$, and write

$$\mathscr{J} = \left\{(J^\ell_q, J^r_q) : 1 \leq q \leq q_0\right\}.$$

Define the selected-block event

$$\mathcal{W}_{\mathscr{J}} := \bigcap_{q=1}^{q_0} \left\{\#\left\{i \in I : |\operatorname{supp} X_i \cap J_q| = 1\right\} \geq c_w d m_*\right\}.$$

By definition, $\mathcal{W}(m_*) \subset \mathcal{W}_{\mathscr{J}}$. It is therefore enough to prove the asserted bounds with $\mathcal{W}_{\mathscr{J}}$ in place of $\mathcal{W}(m_*)$.

On $\mathcal{W}_{\mathscr{J}}$, if

$$b_i = \#\left\{q : |\operatorname{supp} X_i \cap J_q| = 1\right\},$$

then

$$\sum_{i \in I} b_i \geq c_w d m_* q_0. \tag{4.40}$$

In the first case, let I_0 be the witness rows for the sole pair. Then $|I_0| \geq c_w dm_1$ and $b_i = 1$ for every $i \in I_0$. In the second case, use $b_i \leq q_0$ and set

$$a_0 = \frac{c_w}{4}\frac{dm_*}{n}q_0, \qquad I_0 = \{i \in I : b_i \geq a_0\}.$$

Then

$$c_w dm_* q_0 \leq \sum_{i\in I} b_i \leq |I_0|q_0 + na_0,$$

so $|I_0| \geq (3c_w/4)dm_*$. Moreover, $q_0 m_* \geq m_1/2$ gives $a_0 \geq (c_w/8)dm_1/n$. Thus, after adjusting the absolute constant,

$$|I_0| \geq cdm_*, \qquad b_i \geq c\frac{dm_1}{n} \quad (i \in I_0). \tag{4.41}$$

Condition on the counts $Q_{i,q} = |\operatorname{supp} X_i \cap J_q|$ for all $i \in I$ and $q \leq q_0$, on all selected coordinates outside $\bigcup_q J_q$, and on all positions in blocks for which $Q_{i,q} \neq 1$. The local event $\mathcal{W}_{\mathscr{J}}$, the values b_i, and the set I_0 are measurable with respect to this information. Conditional on it, whenever $Q_{i,q} = 1$, the unique coordinate $U_{i,q}$ is uniform on $J_q^\ell \cup J_q^r$; these singleton positions are mutually independent across the disjoint blocks and across rows. By (4.37), a disk of radius $1/3$ cannot contain coefficient values from both halves, so one witnessed block has concentration at most $1/2$. Identifying $\mathbb{C}$ with $\mathbb{R}^2$, the two-dimensional Esseen concentration inequality [LLTTY19, Proposition 2.2] applied to the b_i independent witnessed blocks gives

$$\sup_{\lambda\in\mathbb{C}} \mathbb{P}\left\{\left|\sum_{q:Q_{i,q}=1} u_{U_{i,q}} - \lambda\right| < \frac{1}{3}\right\} \leq \frac{C}{\sqrt{b_i}}, \tag{4.42}$$

with the sharper bound $1/2$ when $b_i = 1$. Every remaining row contribution, including $-\zeta(u_i + w_i)$ and the exact row-sum contribution of w, is conditionally deterministic and merely changes the center.

If the Euclidean norm is below a sufficiently small multiple of $\sqrt{dm_*}$, at least $0.9|I_0|$ retained coordinates have modulus less than $1/3$. Conditional independence across rows and a binomial large-deviation bound give $e^{-c|I_0|}$ in the first case. In the second case, (4.41) implies

$$p_* \leq C\sqrt{\frac{n}{dm_1}},$$

and C_{eb} is chosen sufficiently large that $p_* \leq 1/2$ and $\log(dm_1/n)$ is bounded below by a positive absolute constant. Hence the binomial estimate is

$$\exp[-c|I_0|\log(1/p_*)] \leq \exp\left[-cdm_0 \log\left(\frac{dm_1}{n}\right)\right].$$

Integrating the conditional estimate proves the same bounds with $\mathcal{W}_{\mathscr{J}}$; the inclusion $\mathcal{W}(m_*) \subset \mathcal{W}_{\mathscr{J}}$ then gives (4.38) and (4.39). □

4.7. **The shifted T_1 and T_2 classes.** Normalize the nonconstant component of $T_i^{\mathcal{K}}(L)$ by $x^*_{n_i} = 1$, $i = 1, 2$. The nets in [LLTTY19, Lemma 4.5] have the form

$$\mathcal{N}_i = \mathcal{N}_i' + \mathcal{N}_i'', \qquad \mathcal{N}_i'' \subset \mathcal{K},$$

with

$$|\mathcal{N}_i| \le \exp(7n_{i+1} \log d). \tag{4.43}$$

Every $u \in \mathcal{N}_i'$ is supported on the first n_{i+1} ordered coordinates, and every normalized $x + c\mathbf{1} \in T_i^{\mathcal{K}}(L)$ has an approximant $u + w \in \mathcal{N}_i$ satisfying

$$\|x - u\|_\infty \le d^{-3/2}, \qquad \|c\mathbf{1} - w\|_\infty \le d^{-3/2}. \tag{4.44}$$

In particular,

$$u^*_{n_i} \ge 1 - d^{-3/2} > \frac{2}{3}, \qquad u^*_j = 0 \quad (j \ge n_{i+1}).$$

For the error $h = (x + c\mathbf{1}) - (u + w)$, each row obeys

$$|(A_{\zeta,I}h)_j| \le (d + |\zeta|)\, \|h\|_\infty \le \frac{C}{\sqrt{d}},$$

so

$$\|A_{\zeta,I}h\|_2 \le C\sqrt{\frac{n}{d}}. \tag{4.45}$$

For $T_1^{\mathcal{K}}(L)$, apply lemma 4.2 with $m_1 = n_1$, $m_2 = n_2$. Since $dn_1 \asymp n/\sqrt{d}$ and $\sqrt{n/d} = o(\sqrt{dn_1})$, the net union bound, kept separate from the failure of $\mathcal{W}(n_1)$, gives

$$\begin{aligned}\mathbb{P}\left\{\ker A_{\zeta,I} \cap T_1^{\mathcal{K}}(L) \ne \varnothing\right\} &\le \mathbb{P}(\mathcal{W}(n_1)^c) + \exp(7n_2 \log d - cdn_1) \\ &\le e^{-cn/\sqrt{d}}.\end{aligned} \tag{4.46}$$

Here $n_2 \log d/(dn_1) = O(\log d/d^{1/6}) = o(1)$.

For $T_2^{\mathcal{K}}(L)$, apply the second part of lemma 4.2 with $m_1 = n_2$, $m_2 = n_3$. Then $n_2/m_0 \asymp d^{1/3} \ge C_{\mathrm{eb}}$ for all sufficiently large n, $dm_0 \asymp n$, $dn_2/n \asymp d^{1/3}$, and $\sqrt{n/d} = o(\sqrt{n})$. By the fixed choice of a_3,

$$\begin{aligned}\mathbb{P}\left\{\ker A_{\zeta,I} \cap T_2^{\mathcal{K}}(L) \ne \varnothing\right\} &\le \mathbb{P}(\mathcal{W}(m_0)^c) + \exp(7n_3 \log d - cn \log d) \\ &\le e^{-cn} + e^{-c'n \log d} \le e^{-c''n}.\end{aligned} \tag{4.47}$$

4.8. **The exact almost-constant class.** Set

$$n_0 = \left\lfloor \frac{n}{16d} \right\rfloor, \qquad \mathcal{S}(t) = \left\{x : 0 < x^*_{n_0} \le t x^*_{n_3}\right\}. \tag{4.48}$$

We first record the deterministic reduction used in [LLTTY19, Lemma 4.11].

Lemma 4.3 (Deterministic almost-constant reduction)**.** *For every $L \in \{m, m+1\}$,*

$$(\mathcal{B}(\vartheta_0) \setminus \mathcal{S}(12)) \setminus \{0\} \subset \bigcup_{i=0}^{3} T_i^{\mathcal{K}}(L). \tag{4.49}$$

Proof. Let $0 \neq x \in \mathcal{B}(\vartheta_0) \setminus \mathcal{S}(12)$, and choose λ_0 such that more than $n - n_3$ coordinates satisfy $|x_j - \lambda_0| \le \vartheta_0 x^*_{n_3}$. Put $y = \lambda_0 \mathbf{1}$ and $w = x - y$.

First suppose $x^*_{n_3} > 0$. Then

$$(1 - \vartheta_0)x^*_{n_3} \le |\lambda_0| \le (1 + \vartheta_0)x^*_{n_3}. \tag{4.50}$$

Since $x \notin \mathcal{S}(12)$, one has $x^*_{n_0} > 12x^*_{n_3}$. Therefore

$$w^*_{n_1} \ge w^*_{n_0} \ge x^*_{n_0} - |\lambda_0| > (11 - \vartheta_0)x^*_{n_3},$$

while

$$w^*_{n_3} \le \vartheta_0 x^*_{n_3}.$$

For all sufficiently large d,

$$(11 - \vartheta_0)x^*_{n_3} > 10(1 + \vartheta_0)x^*_{n_3} \ge 10|\lambda_0|,$$

and, because $\vartheta_0 = 10/d^3$,

$$d^3 w^*_{n_3} \le 10x^*_{n_3} < (11 - \vartheta_0)x^*_{n_3} < w^*_{n_1}.$$

If $w \notin T_0(L) \cup T_3(L)$ and also $w \notin T_1(L) \cup T_2(L)$, the definitions would give $w^*_{n_1} \le d^{3/2} w^*_{n_2} \le d^3 w^*_{n_3}$, a contradiction. Thus $w \in T(L)$. Moreover, $|\lambda_0| \le w^*_{n_1}/10$, so $x = w + y$ belongs to the corresponding shifted class.

Now suppose $x^*_{n_3} = 0$. More than $n - n_3$ coordinates of x are zero, and more than $n - n_3$ coordinates equal λ_0; since $a_3 < 1/2$, the two sets intersect and $\lambda_0 = 0$. Thus $w = x$ and $y = 0$. If $w^*_{n_1} > 0$, exclusion from all four source steep classes would imply $w^*_{n_1} \le d^{3/2} w^*_{n_2} \le d^3 w^*_{n_3} = 0$, a contradiction. If $w^*_{n_1} = 0$, then w is supported on fewer than n_1 coordinates. If its support has size below $p_{\mathrm{st}}^{r_0}$, then $w \in T_3(L)$. Otherwise, if the support has size at least $\lceil n_1/p_{\mathrm{st}} \rceil$, the terminal jump in (4.9) is strict because $w^*_{n_1} = 0$. If the support is smaller than $\lceil n_1/p_{\mathrm{st}} \rceil$, there is a first index $i \in \{r_0, \dots, r-1\}$ for which the support lies below p_{st}^{i+1}; the corresponding jump in (4.8) is strict. After accounting for the recursive exclusions, w belongs to the first applicable $T_{0,i}(L)$. Hence $w \in T_0(L)$. In every case $x = w + 0 \in T_i^{\mathcal{K}}(L)$ for some i. □

We next construct the rectangular row-load event. For $J \subset [n]$ and $i \in I$, write

$$H_i(J) = |\operatorname{supp} X_i \cap J|.$$

Let $\mathcal{E}_{\mathrm{load}}$ be the event that simultaneously

$$\#\{i \in I : H_i(J) \ge 1\} \le \frac{K}{8} \qquad \text{for every } |J| \le n_0, \tag{4.51}$$

and

$$\#\{i \in I : H_i(J) \ge 16 n_3 d/n\} \le \frac{K}{8} \qquad \text{for every } |J| \le n_3. \tag{4.52}$$

For fixed J with $|J| \le n_0$, the union bound within one row gives

$$\mathbb{P}\{H_i(J) \ge 1\} \le \frac{d|J|}{n} \le \frac{1}{16}.$$

For fixed J with $|J| \le n_3$, the hypergeometric mean satisfies $\mathbb{E}H_i(J) = d|J|/n \le dn_3/n$, so Markov's inequality gives

$$\mathbb{P}\left\{H_i(J) \ge 16n_3d/n\right\} \le \frac{1}{16}.$$

The rows are independent. A binomial Chernoff bound therefore makes the probability that more than $K/8$ rows are bad at most e^{-cK} for each fixed J. Moreover,

$$\log \sum_{j \le n_0} \binom{n}{j} = O\left(\frac{n \log d}{d}\right) = o(n), \qquad \log \sum_{j \le n_3} \binom{n}{j} \le nH(a_3) + o(n),$$

where $H(t) = -t\log t - (1-t)\log(1-t)$. By the fixed choice of a_3, $H(a_3) < c/2$, and union bounds over all such J give

$$\mathbb{P}(\mathcal{E}_{\text{load}}^c) \le e^{-cn}. \tag{4.53}$$

Take $x \in \mathcal{B}(\vartheta_0) \cap \mathcal{S}(12)$, choose λ_0 as above, and let σ order the coordinates by modulus. Set

$$J_1 = \left\{j : |x_j - \lambda_0| \le \vartheta_0 x^*_{n_3}\right\}, \qquad J_2 = \sigma([n_0]) \setminus J_1,$$
$$J_3 = \sigma([n_3]) \setminus (J_1 \cup J_2), \qquad J_4 = [n] \setminus (J_1 \cup \sigma([n_3])).$$

Then $|J_2| \le n_0$, $|J_3|, |J_4| \le n_3$, and

$$|x_j| \le 12x^*_{n_3} \quad (j \in J_3), \qquad |x_j| \le x^*_{n_3} \quad (j \in J_4).$$

On $\mathcal{E}_{\text{load}}$, delete rows intersecting J_2, rows with load at least $16n_3d/n$ in either J_3 or J_4, and row indices in $\sigma([n_3])$. The surviving set $G \subset I$ has

$$|G| \ge K - \frac{3K}{8} - n_3 \ge \frac{n}{2}. \tag{4.54}$$

For $i \in G$, at least $d - 32n_3d/n$ selected coordinates lie in J_1. Using (4.50), the contribution from J_1 has modulus at least

$$\left(d - \frac{32n_3d}{n}\right)(1 - 2\vartheta_0)x^*_{n_3},$$

whereas the $J_3 \cup J_4$ contribution is at most

$$\frac{208n_3d}{n}x^*_{n_3}.$$

Since $i \notin \sigma([n_3])$, the diagonal term is at most $Z_0\sqrt{d}\,x^*_{n_3}$. By the fixed choice of a_3, and then for all sufficiently large n,

$$|(A_{\zeta,I}x)_i| \ge \frac{d}{2}x^*_{n_3} \qquad (i \in G). \tag{4.55}$$

Thus no vector in $\mathcal{B}(\vartheta_0) \cap \mathcal{S}(12)$ belongs to the kernel on $\mathcal{E}_{\text{load}}$.

Combining lemma 4.3 with (4.32), (4.46), (4.47), and (4.53) yields

$$\mathbb{P}\left\{(\ker A_{\zeta,I} \setminus \{0\}) \cap (\mathcal{B}(\vartheta_0) \setminus T_3^{\mathcal{K}}(L)) \ne \varnothing\right\}$$
$$\le e^{-cn/d^3} + e^{-cn/\sqrt{d}} + e^{-c'n\log d} + e^{-cn}. \tag{4.56}$$

The zero vector is excluded from (4.56): it belongs to $\mathcal{B}(\vartheta_0)$, while the compatibility theorem concerns only nonzero kernel vectors.

4.9. **Compatibility with the exact gradual class.** We can now state the precise theorem needed by the profile argument.

Theorem 4.4 (Rectangular right-kernel compatibility)**.** *There exists an absolute $C_* > 0$, depending only on the fixed constants in the LLTTY decomposition and on a, such that the following holds. Fix $\sigma \in (0,1)$ and $Z_0 < \infty$. Let d satisfy (2.1), let $L \in \{m, m+1\}$, let $I \subset [n]$ be deterministic with $|I| = n - L$, and let $|\zeta| \le Z_0\sqrt{d}$. Then, for all sufficiently large n, where the threshold may depend on σ and Z_0,*

$$\mathbb{P}\left\{\ker A_{\zeta,I} \setminus \{0\} \subset V_L \cup \mathbb{C}^\times S_L\right\} \ge 1 - \frac{C}{n}. \tag{4.57}$$

The constants and the probability bound are uniform for $L = m, m+1$.

Proof. Let $\mathcal{E}_L$ be the event on which

$$\ker A_{\zeta,I} \cap T_i^{\mathcal{K}}(L) = \varnothing, \qquad i = 0,1,2,$$

and

$$(\ker A_{\zeta,I} \setminus \{0\}) \cap (\mathcal{B}(\vartheta_0) \setminus T_3^{\mathcal{K}}(L)) = \varnothing.$$

The estimates above give

$$\mathbb{P}(\mathcal{E}_L^c) \le e^{-cn/d^3} + e^{-cn/\sqrt{d}} + e^{-c'n\log d} + e^{-cn} \le \frac{C}{n}. \tag{4.58}$$

The last inequality is uniform in the lower band because $d \le (\log n)^{1+\sigma}$ implies $n/d^3 \gg \log n$.

Take $0 \ne v \in \ker A_{\zeta,I}$ on $\mathcal{E}_L$. If $v \in T_3^{\mathcal{K}}(L)$, then (4.23) gives $v \in V_L$. Otherwise $v \notin \mathcal{B}(\vartheta_0)$, and $v \notin T(L)$: membership in any unshifted $T_i(L)$ would imply membership in $T_i^{\mathcal{K}}(L)$ by taking the constant vector equal to zero. Moreover, $v^*_{n_3} > 0$, since otherwise $v \in \mathcal{B}(\vartheta_0)$ with center zero. By homogeneity,

$$\frac{v}{v^*_{n_3}} \in S_L.$$

Thus $v \in \mathbb{C}^\times S_L$, proving (4.57). □

Remark 4.5. The final lower bound on C_* must dominate the constants in the support-diameter estimate, the smallest-singular-value theorem, the rectangular restricted-norm estimate, and the witness and net arguments above. It is not introduced solely by the max-weight lemma. The constant C_* is independent of σ and of the fixed shift bound Z_0; only the threshold from which the estimates hold may depend on them, because $Z_0\sqrt{d} = o(d)$.

5. The row-slice structural theorem

We now insert the averaged profile estimate into the deterministic LLTTY decomposition of the exact gradual class S_L. Assume (2.1) and set

$$m = \left\lfloor \frac{n}{d^3} \right\rfloor - 1, \qquad L \in \{m, m+1\}, \qquad K = n - L, \qquad \theta_L = \frac{K}{n}. \tag{5.1}$$

Thus

$$1 - \theta_L = \frac{L}{n} = d^{-3} + O(n^{-1}), \qquad -\log \theta_L = O(d^{-3}). \tag{5.2}$$

The lower-band upper bound gives, uniformly in d,

$$d^2 = o(n), \qquad \frac{n}{d^3} \gg \log n, \qquad d^{1/96} = o(\log n). \tag{5.3}$$

Let $\gamma = 1/288$ and define the exact codimension-dependent parameters

$$u_L = \left(\frac{n}{L}\right)^{\gamma}, \qquad \rho_L = e^{-u_L}, \qquad \delta_L = \left(\frac{L}{n}\right)^{\gamma/2}, \qquad v_L = \left\lfloor \frac{u_L}{\log d} \right\rfloor. \tag{5.4}$$

Then $\rho_L \le d^{-v_L}$ and, uniformly for $L = m, m+1$,

$$\delta_L v_L \asymp \frac{d^{1/192}}{\log d} \gg \log d. \tag{5.5}$$

More explicitly,

$$\delta_L = d^{-1/192 + o(1)}, \qquad v_L \asymp \frac{d^{1/96}}{\log d}.$$

Consequently,

$$c_P a_3 (v_L - 4) \asymp \frac{d^{1/96}}{\log d},$$

whereas

$$2\log_2 d + 2\log_2\left(\frac{72\sqrt{d}}{\delta_L}\right) + 2 - \log_2(c_P a_3) = O(\log d).$$

Thus, for all sufficiently large n,

$$\begin{gathered} 0 < \delta_L \le 36 c_P a_3, \\ c_P a_3 (v_L - 4) \ge 2\log_2 d + 2\log_2\left(\frac{72\sqrt{d}}{\delta_L}\right) + 2 - \log_2(c_P a_3). \end{gathered} \tag{5.6}$$

Hence the hypotheses in part (v) of proposition 2.2 hold at $(v, \rho, \delta) = (v_L, \rho_L, \delta_L)$. By (5.3),

$$d^{v_L} = \exp(O(d^{1/96})) = n^{o(1)}, \qquad \frac{\sqrt{n}}{d^{3/2}} = n^{1/2 - o(1)}.$$

In particular, $d \le n^{1/3}$ and $d^{v_L} \le \sqrt{n}/d^{3/2}$ for all sufficiently large n, so all LLTTY net and equivalence-class bounds apply for $4 \le u \le v_L$.

For later reference, define the clustered terminal class

$$P_{v_L, \rho_L, \delta_L}(L) := \left\{ x \in P_{v_L}(L) : \exists \lambda \in \mathbb{C},\ \#\left\{ j : |x_j - \lambda| \le \rho_L x^*_{n_3} \right\} > \delta_L n \right\}. \tag{5.7}$$

5.1. **The net reduction.** The following is the rectangular form of the deterministic net reduction in [LLTTY19, Sections 5–7]. The only matrix estimate needed in passing from a continuous vector to its net point is the rectangular restricted norm from proposition 2.6.

Lemma 5.1 (Row-slice net reduction)**.** *Let $\mathcal{X}$ be one of the families $K_u(L)$ or $P_{v_L,\rho_L,\delta_L}(L)$, and let $k = d^u$ be its discretization scale. There is a deterministic set $\mathcal{A}_k(\mathcal{X})$ of k-vectors such that*

$$\mathbb{P}\{\exists x \in \mathcal{X} : A_{\zeta,I}x = 0\} \le 2d^2 \sum_{y\in\mathcal{A}_k(\mathcal{X})} \sup_{V\in\mathbb{C}^K} \times \mathbb{P}\left\{\|M_I y + V\|_2 \le C\log d\frac{\sqrt{dn}}{k}\right\} + n^{-100}. \tag{5.8}$$

The k-vectors represented in $\mathcal{A}_k(\mathcal{X})$ fall into at most e^n LLTTY equivalence classes, and each represented class obeys (2.8). Moreover,

$$\left|\bigcup_{u=4}^{v_L} \mathcal{A}_{d^u}(L)\right| \le v_L \exp(Cnd^{1/3}). \tag{5.9}$$

Lemma 5.1A (Coordinate ranges for gradual net points). Let $x \in S_L$, so $x^*_{n_3} = 1$, and let y be its canonical k-approximation. If σ orders the coordinates of x by nonincreasing modulus, then, for all sufficiently large n,

$$\max_{j\le n}|y_j| \le 2n^6, \qquad \max_{j>n_2}|y_{\sigma(j)}| \le 2d^{3/2}.$$

Proof of Lemma 5.1A. Because $x \in S_L \subset T(L)^c$ and $x^*_{n_3} = 1$, the deterministic order-statistic estimate in [LLTTY19, Lemma 4.2] gives $x^*_1 \le n^6 x^*_{n_3} = n^6$. Since $x \notin T_2(L)$, the defining reverse inequality gives $x^*_{n_2} \le d^{3/2}x^*_{n_3} = d^{3/2}$. Canonical rounding changes each real and imaginary component by less than $1/k$, and $k \ge 1$, so the displayed bounds follow after replacing the constants by 2. □

Proof of lemma 5.1. Work on the restricted-norm event from proposition 2.6. Let $x \in \mathcal{X}$ satisfy $A_{\zeta,I}x = 0$, and let y be its canonical k-approximation. Put

$$\alpha = \frac{1}{n}\sum_{j=1}^{n}(x_j - y_j), \qquad w = x - y - \alpha\mathbf{1}.$$

Then $w \perp \mathbf{1}$, while coordinatewise rounding gives

$$\|w\|_2 \le C\frac{\sqrt{n}}{k}, \qquad |\alpha| \le \frac{C}{k}. \tag{5.10}$$

Round the real and imaginary parts of α to the grid $(kd)^{-1}\mathbb{Z}$, obtaining $\widetilde{\alpha}$. There are at most Cd^2 possible values and

$$|\alpha - \widetilde{\alpha}| \le \frac{C}{kd}. \tag{5.11}$$

Writing $y_I = (y_i)_{i\in I}$, define

$$V(y,\widetilde{\alpha}) = -\zeta y_I + \widetilde{\alpha}(d-\zeta)\mathbf{1}_I.$$

The exact row-sum identity $M_I\mathbf{1} = d\mathbf{1}_I$ and $A_{\zeta,I}x = 0$ give

$$M_I y + V(y,\widetilde{\alpha}) = -(M_I - \zeta P_I)w + (\widetilde{\alpha}-\alpha)(d-\zeta)\mathbf{1}_I.$$

Since $w \perp \mathbf{1}$, one has $M_I w = (M_I - \mathbb{E}M_I)w$. Therefore (5.10), (5.11), and the restricted norm imply

$$\|M_I y + V(y,\widetilde{\alpha})\|_2 \le C_{Z_0}\frac{\sqrt{dn}}{k} \le C\log d\frac{\sqrt{dn}}{k}.$$

Taking the union over $O(d^2)$ choices of $\widetilde{\alpha}$ proves (5.8). We use proposition 2.6 with failure n^{-100}.

The equivalence-class assertions for the canonical approximations in $\mathcal{A}_k(\mathcal{X})$ are [LLTTY19, Lemmas 5.1 and 5.5]; no claim is made here about unrestricted k-vectors. For the crude count, first choose the positions of the top n_2 coordinates. Lemma 5.1A shows that each top coordinate has $O(k^2n^{12})$ possible complex grid values, while each remaining coordinate has $O(k^2d^3)$ possible values. Hence

$$|\mathcal{A}_k(L)| \le \binom{n}{n_2}(Ck^2n^{12})^{n_2}(Ck^2d^3)^{n-n_2}.$$

Since $n_2 \asymp n/d^{2/3}$ and $\log k = O(d^{1/96})$, the logarithm of the right-hand side is at most $Cnd^{1/3}$. Summing over the v_L discretization scales proves (5.9). □

For a fixed net vector y, define W_y and B by (3.8) and (3.7). Applying (3.12) and (3.13) gives

$$\mathbb{P}_{\text{slice}}\left\{\|M_I y + V\|_2 \le C\log d\frac{\sqrt{dn}}{k}\right\} \le e^{Cd^2}\left[\Lambda_L\big(Ce^{-cD(B)}W_y + e^{-cd^{2/3}}\big)\right]^K, \tag{5.12}$$

where

$$\Lambda_L = C\frac{n}{K}\log^2 d = C\theta_L^{-1}\log^2 d. \tag{5.13}$$

We split the two terms using $(A+B)^K \le 2^{K-1}(A^K + B^K)$.

5.2. **The profile-exception term.** By (5.9), the additive term $e^{-cd^{2/3}}$, summed over all $4 \le u \le v_L$, contributes at most

$$\exp\left[-cKd^{2/3} + Cnd^{1/3} + CK\log\log d + Cd^2 + O(n)\right] \le e^{-c'nd^{2/3}}.$$

This single estimate is used for every branch below.

5.3. **Entropy of dyadic weight orders.**

Lemma 5.2. *For every integer-valued random variable B,*

$$H(B) \le C + \log(1 + D(B)), \qquad D(B) = \mathbb{E}|B - B'|.$$

Proof. Let $r \in \mathbb{Z}$ be a median of B. Since a median minimizes expected absolute deviation,

$$M := \mathbb{E}|B - r| \le \mathbb{E}|B - B'| = D(B).$$

For $\alpha \in (0, 1]$, compare the law of B by relative entropy with

$$q_j = Z_\alpha^{-1} e^{-\alpha|j-r|}, \qquad Z_\alpha \le C/\alpha.$$

Then $H(B) \le \log Z_\alpha + \alpha M$. Taking $\alpha = (1+D(B))^{-1}$ proves the lemma. □

Let T_b be the dyadic weight set of order b, and let $t_b = |T_b|$. By (2.9),

$$\frac{n!}{\prod_q s_q!} \le e^{Cn} \frac{n!}{\prod_b t_b!} \le e^{Cn + nH(B)}. \tag{5.14}$$

Combining lemma 5.2 with $K/n \to 1$ yields

$$\frac{n!}{\prod_q s_q!} e^{-cKD(B)} \le e^{C'n}, \tag{5.15}$$

because $\log(1 + t) - c(K/n)t$ is uniformly bounded above.

5.4. **The $K_u(L)$ branches.** Let y be the d^u-approximation of a vector in $K_u(L)$, and put

$$\kappa = \frac{c_K a_3}{16} < \frac{1}{3}, \qquad \mathscr{I} = \{q : s_q \ge d^{-\kappa} n\}. \tag{5.16}$$

The inequality $\kappa < 1/3$ ensures that every part in $\mathscr{I}$ uses the large-part branch of (2.4).

5.4.1. *Case 1: large parts occupy almost all coordinates.* Assume

$$\sum_{q \in \mathscr{I}} s_q \ge n - \frac{c_K n_3}{4}.$$

Let $\mathscr{I}_1$ be the large spread parts, $\mathscr{I}_2$ the large regular parts, and $\mathscr{I}_3$ the remaining parts. Then

$$\sum_{q \in \mathscr{I}_1} s_q \ge \frac{3c_K n_3}{4}, \qquad \sum_{q \in \mathscr{I}_3} s_q \le \frac{c_K n_3}{4}.$$

For $q \in \mathscr{I}_1$,

$$\widetilde{w}_q^{-s_q} \le d^{-s_q/2} \left(\frac{n}{s_q}\right)^{s_q} h_q^{-s_q};$$

for $q \in \mathscr{I}_2$,

$$\widetilde{w}_q^{-s_q} = \left(\frac{n}{s_q}\right)^{s_q} h_q^{-s_q};$$

and for $q \in \mathscr{I}_3$,

$$\widetilde{w}_q^{-s_q} \le d^{s_q} h_q^{-s_q}.$$

Since $n/s_q \le d^\kappa$ on $\mathscr{I}$, multiplication gives

$$\prod_q \widetilde{w}_q^{-s_q} \le C^n d^{-c_K n_3/16} \prod_q h_q^{-s_q}.$$

Thus, for some absolute $c_1 > 0$,

$$\prod_q h_q^{s_q} W_y^n \le C^n d^{-c_1 n}. \tag{5.17}$$

Passing from n to $K = n - L$ rows costs only

$$W_y^K = W_y^n W_y^{-L}, \qquad W_y^{-L} \le (n\sqrt{d})^L = e^{O(n/d^2)}. \tag{5.18}$$

For one equivalence class, multiply the main term in (5.12) by (2.8). Using (5.15), (5.17), and (5.18), its logarithm is

$$-c_1 n \log d + O(n \log\log d + n + d^2 + n/d^2).$$

The e^n equivalence classes and v_L choices of u are absorbed, so

$$\mathbb{P}\{\text{some Case-1 } K_u(L) \text{ net vector is small}\} \le e^{-cn\log d}. \tag{5.19}$$

5.4.2. *Case 2: a constant fraction lies in small parts.* Assume

$$\sum_{q \in \mathscr{I}^c} s_q \ge \frac{c_K n_3}{4}.$$

By proposition 2.2, there are $b_1 \le b_2$ such that

$$\mathbb{P}\{B \le b_1\} \ge c_0, \qquad \mathbb{P}\{B \ge b_2\} \ge c_0, \qquad b_2 - b_1 \ge c_1 d^\kappa.$$

Therefore

$$D(B) \ge 2c_0^2(b_2 - b_1) \ge c d^\kappa. \tag{5.20}$$

Using $\widetilde{w}_q \ge h_q/d$,

$$W_y^K \prod_q h_q^{s_q} \le d^K \prod_q h_q^{(1-K/n)s_q} \le d^K e^{O(n/d^2)}.$$

The factorial factor is at most $(3\log n)^n = e^{O(n\log d)}$. Thus the contribution of one class has logarithm

$$-cKd^\kappa + O(n\log d + n/d^2 + d^2).$$

Since $d^\kappa \gg \log d$,

$$\mathbb{P}\{\text{some Case-2 } K_u(L) \text{ net vector is small}\} \le e^{-cnd^\kappa}. \tag{5.21}$$

Combining the two cases and the profile-exception term gives

$$\sum_{u=4}^{v_L} \sum_{y \in \mathcal{A}_{d^u}(K_u(L))} \sup_V \mathbb{P}\left\{\|M_I y + V\|_2 \le C\log d \frac{\sqrt{dn}}{d^u}\right\} \le e^{-cn\log d}. \tag{5.22}$$

5.5. **The clustered $P_{v_L}(L)$ branch.** Let $x \in P_{v_L,\rho_L,\delta_L}(L)$, and let y be its d^{v_L}-approximation. By part (v) of proposition 2.2, with the hypotheses verified in (5.6), there exist $b_1 < b_2$ such that

$$\mathbb{P}\{B \le b_1\} \ge \frac{\delta_L}{36}, \qquad \mathbb{P}\{B \ge b_2\} \ge c_0, \qquad b_2 - b_1 \ge \frac{c_P a_3}{2}(v_L - 4) \ge c v_L.$$

Consequently,

$$D(B) \ge c\delta_L v_L. \tag{5.23}$$

For one equivalence class, use $\widetilde{w}_q \ge h_q/d$ in the main term. Multiplication by (2.8) leaves

$$e^{O(d^2)}(Cd\log^2 d)^K e^{-cK\delta_L v_L}\frac{n!}{\prod_q s_q!}\prod_q h_q^{(1-K/n)s_q}.$$

The factorial factor is $e^{O(n\log d)}$, the height mismatch is $e^{O(n/d^2)}$, and $\delta_L v_L \gg \log d$. Therefore

$$\mathbb{P}\{\text{some clustered } P_{v_L}(L) \text{ net vector is small}\} \le e^{-cn\delta_L v_L} + e^{-cnd^{2/3}}. \tag{5.24}$$

5.6. **Assembly.** By proposition 2.2, the exact gradual class decomposes as

$$S_L = \bigcup_{u=4}^{v_L} K_u(L) \cup P_{v_L}(L). \tag{5.25}$$

The estimates above show that, with failure $o(n^{-1})$, no vector in $\mathbb{C}^\times S_L$ lying in the right kernel has a cluster of more than $\delta_L n$ coordinates at radius $\rho_L x^*_{n_3}$. The compatibility theorem from theorem 4.4 identifies every remaining nonzero kernel vector with either V_L or $\mathbb{C}^\times S_L$.

Theorem 5.3 (Row-slice structural theorem)**.** *There are absolute constants $a, C_* > 0$ such that the following holds. Fix $\sigma \in (0,1)$, let d satisfy (2.1), and set*

$$m = \left\lfloor n/d^3 \right\rfloor - 1, \qquad L \in \{m, m+1\}.$$

Let $I \subset [n]$ be deterministic with $|I| = n - L$. Put $q_L = \lfloor aL \rfloor$, and let

$$E_I = \operatorname{span}\left\{\overline{X_i - \zeta e_i} : i \in I\right\}, \qquad |\zeta| \le Z_0\sqrt{d}.$$

Then, with probability at least $1 - C/n$, every nonzero $x \in E_I^\perp = \ker A_{\zeta,I}$ satisfies one of the following:

(i) *there exists $i \le q_L$ such that*

$$x_i^* > 0.9\left(\frac{n}{i}\right)^3 x^*_{q_L};$$

(ii) *for all $i \le q_L$ the reverse inequality holds, and*

$$\sup_{\lambda\in\mathbb{C}} \#\left\{j : |x_j - \lambda| \le \exp\left[-2\left(\frac{n}{L}\right)^{1/288}\right] x^*_{q_L}\right\} \le n\left(\frac{L}{n}\right)^{1/576}.$$

The statement is for each fixed I, not simultaneously over all row sets. Its constants are uniform in d throughout (2.1) and for $L = m, m+1$; the threshold from which it holds may depend on σ and Z_0.

Proof. By theorem 4.4, outside an event of probability C/n, every nonzero kernel vector is in V_L or in $\mathbb{C}^\times S_L$. The first class is alternative (i). Consider a vector in the second class that is not in V_L. The deterministic order-statistic estimates built into the LLTTY gradual class [LLTTY19, Lemma 4.2] give

$$x^*_{q_L} \le d^3 \left(\frac{n}{q_L}\right)^6 x^*_{n_3} \le C_a d^{21} x^*_{n_3}. \tag{5.26}$$

Since $u_L/\log d \to \infty$,

$$C_a d^{21} \le e^{u_L}$$

for all large n. Hence

$$e^{-2u_L} x^*_{q_L} \le e^{-u_L} x^*_{n_3} = \rho_L x^*_{n_3}. \tag{5.27}$$

A cluster forbidden by alternative (ii) would therefore be a cluster in $P_{v_L,\rho_L,\delta_L}(L)$, which is excluded by (5.22), (5.24), and lemma 5.1 with failure $o(n^{-1})$. Since $\delta_L n = n(L/n)^{1/576}$, the stated bound follows. □

6. Gaussian normals and exact-slice anti-concentration

The structural theorem is converted into anti-concentration for a random normal. This stage is model-independent once the row span has the structural property.

6.1. **The Gaussian cluster theorem.** Let $E \subset \mathbb{C}^n$ be a subspace and let G be a standard complex Gaussian vector. We write

$$Y = P_{E^\perp} G.$$

We use the following fixed measurable enlargement rule. If a subspace $F \subset \mathbb{C}^n$ has dimension below a prescribed target r, set $F_0 = F$ and recursively choose the least index j for which $P_{F_{k-1}^\perp} e_j \neq 0$, normalize that projection, and adjoin it to F_{k-1}. Stop at dimension r, and denote the resulting subspace by $\mathrm{Ext}_r(F)$. Orthogonal projection, the least-index choice, and Gram–Schmidt normalization are Borel measurable on each rank stratum, so $F \mapsto \mathrm{Ext}_r(F)$ is measurable. It contains F. Consequently, if $F \subset E_0$, then $E_0^\perp \subset F^\perp$, every structural statement valid on $F^\perp$ remains valid on $E_0^\perp$, and

$$\mathrm{dist}(R, E_0) \le \mathrm{dist}(R, F)$$

for every row R. Thus rank deficiency creates no obstruction.

The following is the fixed-subspace content used from [LLTTY21, Theorem 4.2 and Lemmas 4.8–4.10]. More precisely, [LLTTY21, Lemma 4.10] supplies the deterministic implication from an oversized correlation-cluster exceptional set to a forbidden vector in $E^\perp$, while [LLTTY21, Lemmas 4.8 and 4.9] supply the Gaussian anti-clustering estimate after the large clusters have been removed. The proposition below isolates only this conditional

fixed-subspace interface: E is deterministic throughout, and all probability is over G. No regular-digraph probability estimate is imported at this stage.

Proposition 6.1 (Fixed-subspace Gaussian anti-clustering)**.** *Let* $\gamma = 1/288$ *and*

$$\frac{n}{\log^{1/\gamma} n} \le L \le \frac{n}{d^3}, \qquad q_L = \lfloor aL \rfloor, \qquad a \le c_{4.10}.$$

Let $E \subset \mathbb{C}^n$ *be deterministic with* $\dim E = n - L$*. Suppose that every nonzero* $x \in E^\perp$ *satisfies at least one of the following exact alternatives:*

(i) *there is* $i \le q_L$ *such that*

$$x_i^* > 0.9 \left(\frac{n}{i}\right)^3 x_{q_L}^*;$$

(ii) *for every* $i \le q_L$ *the reverse inequality holds, and*

$$\sup_{\lambda \in \mathbb{C}} \# \left\{ j : |x_j - \lambda| \le e^{-2(n/L)^\gamma} x_{q_L}^* \right\} \le n \left(\frac{L}{n}\right)^{\gamma/2}.$$

Let G *be a standard complex Gaussian vector and* $Y = P_{E^\perp} G$*. Then, with probability at least* $1 - CL/n$ *over* G*, there is* $J \subset [n]$ *such that*

$$|J| \le 2n \left(\frac{L}{n}\right)^{\gamma/2} \tag{6.1}$$

and

$$\sup_{\lambda \in \mathbb{C}} \# \left\{ j \notin J : |Y_j - \lambda| \le e^{-C(n/L)^\gamma} \right\} \le L. \tag{6.2}$$

Proof. Only the fixed-subspace Gaussian results of [LLTTY21] are used. The notation and hypotheses are mapped as follows.

Source quantity	**Choice here**	**Verification**
$m = \dim E^\perp$	$m = L$	$L \ge n/\log^{288} n \ge n^{3/4}$ for large n
number of Gaussian copies	$p_{\mathrm{src}} = \ell_0 - 1$	$p_{\mathrm{src}} < n/b \le 3n/L \le n^{1/4}$ for large n
cluster cutoff b	$b = \lfloor L/2 \rfloor$	$1 \le b < L$ and $2b \le L$
small-ball parameter s	$s = \frac{1}{2}(L/n)^{3/2}$	positive and deterministic
strong-correlation radius α	$e^{-C_G(n/L)^\gamma}$	the radius used in the source construction
correlation probability β	$\frac{1}{4}(L/(4n))^3$	the source Lemma 4.10 scale
exceptional-size cutoff V	$2n(L/n)^{\gamma/2}$	the source Lemma 4.10 scale
Lemma 4.10 index ℓ_0	$p_{\mathrm{src}} + 1$	$\ell_0 < n/b + 1 \le 4n/L$ for large L
source steep cutoff	$\lfloor c_{4.10} L \rfloor$	$q_L \le \lfloor c_{4.10} L \rfloor$

The hypotheses of [LLTTY21, Propositions 4.4 and 4.7] are therefore both satisfied by the displayed stronger bound $p_{\mathrm{src}} \le n^{1/4}$; the dimension requirement $m \ge n^{1/2}$ follows from $L \ge n^{3/4}$.

Call a pair (i,j) (α,β)-strongly correlated if

$$\mathbb{P}_G\{|Y_i - Y_j| \ge \alpha\} \le \beta.$$

Fix once and for all a total order on the subsets of $[n]$ that first compares cardinality and then compares indicator vectors lexicographically. At each stage of the greedy construction, choose the greatest admissible cluster in this order. This is the deterministic tie-break used immediately before [LLTTY21, Lemma 4.8]; it makes every cluster U_ℓ and the exceptional set below measurable functions of the covariance of Y.

Construct the disjoint strong-correlation clusters $U_1, U_2, \dots$ with this tie-break. Let ℓ_0 be the first index with $|U_{\ell_0}| \le b$, and put

$$J = \bigcup_{\ell < \ell_0} U_\ell.$$

If $|J| > V$, [LLTTY21, Lemma 4.10] produces a nonzero $x \in E^\perp$ satisfying the reverse steepness inequalities through $\lfloor c_{4.10} L \rfloor$ and having more than $n(L/n)^{\gamma/2}$ coordinates in a disk of radius

$$e^{-2(n/L)^\gamma} x^*_{\lfloor c_{4.10} L \rfloor}.$$

Because $q_L \le \lfloor c_{4.10} L \rfloor$, order-statistic monotonicity gives

$$x^*_{\lfloor c_{4.10} L \rfloor} \le x^*_{q_L}.$$

The vector is therefore neither in alternative (i) nor in alternative (ii), a contradiction. Hence $|J| \le V$ under the structural hypothesis.

Outside J, all greedy clusters have size at most b. Apply [LLTTY21, Lemmas 4.8 and 4.9] with the displayed α, β, b, s. The probability of a disk containing at least $2b$ coordinates outside J at radius

$$r_G = \frac{\alpha s}{\sqrt{16 \log(1/\beta)}}$$

is at most

$$\frac{(sn)^2}{2b^2} \le C \frac{L}{n}.$$

Since $2b \le L$, absence of such a disk implies the cardinality bound in (6.2). Finally, writing $q = n/L \ge 2$,

$$\log r_G = -C_G q^\gamma - \frac{3}{2} \log q - O(\log(e + \log q)) \ge -C q^\gamma,$$

so $r_G \ge e^{-C(n/L)^\gamma}$. Anti-clustering at radius r_G implies the claim at the smaller displayed radius. □

Under (2.1),

$$L \ge m \ge \frac{n}{2d^3} \ge \frac{n}{\log^{3(1+\sigma)+1} n} \ge \frac{n}{\log^{288} n}$$

for all sufficiently large n, uniformly for $L = m, m+1$. Thus the lower codimension hypothesis in proposition 6.1 is satisfied throughout the lower band. Combining theorem 5.3 and proposition 6.1 with $L \in \{m, m+1\}$ and $m = n/d^3 + O(1)$ gives

$$\rho := e^{-Cd^{1/96}}, \qquad |J| \le 2nd^{-1/192+o(1)}, \tag{6.3}$$

and failure probability $O(L/n) + O(n^{-1}) = O(d^{-3}) + O(n^{-1})$.

6.2. **Quotienting the constant direction.** For $y \in \mathbb{C}^n$, let

$$\mathfrak{m}(y) = \frac{1}{n}\sum_{j=1}^{n} y_j, \qquad y^\circ = y - \mathfrak{m}(y)\mathbf{1}.$$

If X is a uniform d-subset row, then

$$X \cdot y = d\mathfrak{m}(y) + X \cdot y^\circ.$$

The deterministic term $d\mathfrak{m}(y)$ changes only the center in the Lévy concentration function. For a shifted row $R_i = X - \zeta e_i$,

$$R_i \cdot y = X \cdot y^\circ + d\mathfrak{m}(y) - \zeta y_i,$$

and the last two terms are deterministic after conditioning on y and i. Coordinate clusters are likewise invariant under subtracting $\mathfrak{m}(y)$. Thus all small-ball estimates are naturally statements modulo $\operatorname{span}\{\mathbf{1}\}$.

6.3. **A double-counting lemma for an exact row slice.**

Lemma 6.2 (Slice anti-concentration from coordinate anti-clustering)**.** *Let $J \subset [n]$, $K = J^c$, and suppose $|K| \ge n/2$. Assume that for some $m \ge 1$ and $\rho > 0$,*

$$\sup_{\lambda\in\mathbb{C}} \#\{j \in K : |y_j - \lambda| \le \rho\} \le m. \tag{6.4}$$

If X is the indicator of a uniform d-subset of $[n]$ and $d \le n/2$, then

$$\sup_{\lambda\in\mathbb{C}} \mathbb{P}\{|X \cdot y - \lambda| \le \rho\} \le \frac{4m}{n-d+1} + e^{-cd}. \tag{6.5}$$

Proof. Let S be the random d-subset and put $H(S) = |S \cap K|$. Since $|K| \ge n/2$, hypergeometric concentration gives

$$\mathbb{P}\{H < d/4\} \le e^{-cd}. \tag{6.6}$$

Fix $\lambda \in \mathbb{C}$ and let

$$\mathcal{A} = \left\{ S : |S| = d,\ \left|\sum_{j\in S} y_j - \lambda\right| \le \rho \right\}.$$

Double-count pairs (S, t) with $S \in \mathcal{A}$ and $t \in S \cap K$:

$$\sum_{|S|=d} \mathbf{1}_{\mathcal{A}}(S)H(S) = \sum_{|T|=d-1} \sum_{\substack{t\in K\\ t\notin T}} \mathbf{1}\left\{ \left| y_t + \sum_{j\in T} y_j - \lambda \right| \le \rho \right\}.$$

For each fixed T, the inner sum is at most m by (6.4). Therefore

$$\sum_{|S|=d} \mathbf{1}_{\mathcal{A}}(S) H(S) \le m \binom{n}{d-1}.$$

On $H(S) \ge d/4$,

$$\frac{d}{4} \# \{S \in \mathcal{A} : H(S) \ge d/4\} \le m \binom{n}{d-1}.$$

Dividing by $\binom{n}{d}$ and using (6.6) gives

$$\mathbb{P}(\mathcal{A}) \le \frac{4m}{d} \frac{\binom{n}{d-1}}{\binom{n}{d}} + e^{-cd} = \frac{4m}{n-d+1} + e^{-cd}.$$

Taking the supremum over λ proves the lemma. □

At $m = n/d^3 + O(1)$, the right-hand side of (6.5) is $O(d^{-3})$.

6.4. **The averaged random-normal small-ball bound.** Fix a deterministic exposed row set I of one of the two required cardinalities, and let

$$F_I = \operatorname{span}\left\{\overline{X_i - \zeta e_i} : i \in I\right\}$$

be the actual conjugated row span. Define

$$E_I = \operatorname{Ext}_{|I|}(F_I), \qquad Y = P_{E_I^\perp} G,$$

where G is an independent standard complex Gaussian. Let

$$\mathcal{F}_I = \sigma\big((X_i)_{i \in I}, G\big).$$

Then E_I, Y, the greedy clusters, and the exceptional set J are $\mathcal{F}_I$-measurable. By (4.1), $E_I^\perp \subset F_I^\perp = \ker A_{\zeta,I}$, so Y is a Gaussian vector in the right kernel of the exposed shifted matrix. Conjugation is an isometry and preserves coordinate-cluster cardinalities.

A fresh row slice X is independent of $\mathcal{F}_I$. Conditional on $\mathcal{F}_I$, theorem 5.3 and proposition 6.1 give coordinate anti-clustering outside J, and lemma 6.2 therefore yields

$$\mathbb{E}\left[\sup_{\lambda \in \mathbb{C}} \mathbb{P}_X\left\{|X \cdot Y - \lambda| \le e^{-Cd^{1/96}} \,\middle|\, \mathcal{F}_I\right\}\right] \le Cd^{-3} + \frac{C}{n}. \tag{6.7}$$

The structural and Gaussian failure events contribute $O(n^{-1}) + O(d^{-3})$, and the concentration function is at most one. The exceptional coordinate set from (6.3) has size $o(n)$, hence its complement has size at least $n/2$ for large n. The estimate is uniform for each deterministic fixed I of the two required cardinalities. No simultaneous event over all choices of I is asserted.

7. Intermediate singular values

We convert (6.7) into a lower bound for an intermediate singular value. Fix $z \in \mathbb{C}$ and write

$$A_z = M_n - z\sqrt{d}\, I_n.$$

Let $N = n - m$ with

$$m = \left\lfloor n/d^3 \right\rfloor - 1,$$

and expose the first N rows $R_1, \dots, R_N$ of A_z.

For each $i \le N$, let

$$F_i = \operatorname{span}\left\{\overline{R_j} : j \le N,\ j \ne i\right\}$$

and define the actual leave-one-row-out distance

$$D_i = \operatorname{dist}(\overline{R_i}, F_i) = \operatorname{dist}\big(R_i, \operatorname{span}\{R_j : j \le N,\ j \ne i\}\big).$$

If $\dim F_i < N - 1$, set $E_i = \operatorname{Ext}_{N-1}(F_i)$; otherwise set $E_i = F_i$. Put

$$\widetilde{D}_i = \operatorname{dist}(\overline{R_i}, E_i) \le D_i, \qquad Y_i = P_{E_i^\perp} G_i,$$

where G_i is an independent standard complex Gaussian. Conditional on R_i and E_i,

$$R_i \cdot Y_i = \left\langle Y_i, \overline{R_i} \right\rangle = \left\langle G_i, P_{E_i^\perp} \overline{R_i} \right\rangle$$

is a centered complex Gaussian with variance $\widetilde{D}_i^2$. Hence there are absolute $c, c_0 > 0$ such that

$$\widetilde{D}_i \le c\rho \quad \Longrightarrow \quad \mathbb{P}_{G_i}\left\{|R_i \cdot Y_i| \le \rho \,|\, R_i, E_i\right\} \ge c_0, \tag{7.1}$$

where

$$\rho = e^{-C_{\mathrm{isv}}(z) d^{1/96}}$$

for a sufficiently large shift-dependent constant $C_{\mathrm{isv}}(z)$.

Conditional on E_i, Y_i, the omitted row $R_i = X_i - z\sqrt{d}\, e_i$ remains an independent row slice. The diagonal shift changes only the center of the concentration function, so (6.7) gives

$$\mathbb{P}\left\{|R_i \cdot Y_i| \le \rho\right\} \le C d^{-3} + C/n. \tag{7.2}$$

Combining (7.1) and (7.2),

$$\mathbb{P}\left\{D_i \le c\rho\right\} \le \mathbb{P}\left\{\widetilde{D}_i \le c\rho\right\} \le C d^{-3} + C/n. \tag{7.3}$$

For each fixed index i, the exposed set $\{1, \dots, N\} \setminus \{i\}$ is deterministic, so the fixed-I estimate in Section 6 applies. We do not intersect these N structural events and do not take a union bound over the leave-one-out indices. Instead, the per-index bound is summed only after taking expectations.

Let

$$B = \#\left\{i \le N : D_i < c\rho\right\}.$$

Linearity of expectation, not independence, gives

$$\mathbb{E}B \le C\frac{n}{d^3} + C.$$

Set

$$t = \left\lceil \frac{n}{d^2} \right\rceil.$$

Markov's inequality yields

$$\mathbb{P}\{B > t\} = O\left(\frac{1}{d} + \frac{d^2}{n}\right) = O(1/d) = o(1). \tag{7.4}$$

On $B \le t$, delete all bad rows and, if necessary, additional arbitrary rows so that the resulting matrix H has $k = N - t$ rows. For every surviving row h_i,

$$\operatorname{dist}(h_i, \operatorname{span}\{h_j : j \neq i\}) \ge c\rho,$$

because deleting rows only shrinks the comparison span. By the negative second-moment identity (2.17),

$$\sum_{j=1}^{k} s_j(H)^{-2} \le \frac{k}{c^2\rho^2}. \tag{7.5}$$

Let $j_0 = N - 2t = k - t$. At least $t + 1$ singular values of H are no larger than $s_{j_0}(H)$, and therefore

$$\frac{t+1}{s_{j_0}(H)^2} \le \sum_{j=1}^{k} s_j(H)^{-2}.$$

Using (7.5),

$$s_{j_0}(H) \ge c\rho\sqrt{\frac{t+1}{k}} \ge \frac{c\rho}{d}. \tag{7.6}$$

Adding the deleted rows replaces the common $n \times n$ column Gram matrix H^*H by

$$H^*H + \sum_{u \in \mathcal{D}} R_u^* R_u,$$

a positive-semidefinite increment. Hence the relevant singular values can only increase, and

$$s_{j_0}(A_z) \ge s_{j_0}(H).$$

Define

$$q_0 = m + 2t.$$

Then $j_0 = n - q_0$ and

$$q_0 = O\left(\frac{n}{d^2}\right) = o\left(\frac{n}{d \log d}\right). \tag{7.7}$$

Since the polynomial factor d^{-1} is absorbed into the exponential, (7.6) proves the desired estimate.

Theorem 7.1 (Lower-band intermediate singular value)**.** *Fix $\sigma \in (0,1)$ and let d satisfy (2.1). Let M_n be the independent row-slice matrix. For every fixed $z \in \mathbb{C}$, there are constants $C_{\mathrm{isv}}(z), C'_{\mathrm{isv}}(z) > 0$ and integers $q_0 = q_0(n,z)$ with*

$$q_0 \le C'_{\mathrm{isv}}(z)\frac{n}{d^2}$$

such that

$$\mathbb{P}\left\{s_{n-q_0}(M_n - z\sqrt{d}\, I_n) \geq e^{-C_{\mathrm{isv}}(z)d^{1/96}}\right\} = 1 - o(1).$$

More precisely, the failure probability is $O(1/d)$ uniformly throughout (2.1); the term d^2/n and the exponentially small net and witness errors are absorbed into this bound.

Remark 7.2. The choice

$$m = \left\lfloor n/d^3 \right\rfloor - 1 = \frac{n}{d^3} + O(1)$$

is tuned to the LLTTY exponent $\gamma = 1/288$:

$$(n/m)^{\gamma} = d^{1/96+o(1)}, \qquad (m/n)^{\gamma/2} = d^{-1/192+o(1)}.$$

The deletion scale $t = n/d^2$ is larger than the expected number n/d^3 of bad rows, but remains small enough for the lower-band logarithmic-tail budget.

8. Generalized regularized row replacement

We prove the comparison estimate used in both the lower-band and superlogarithmic regimes. The argument is a regularized version of the prefix-volume replacement in [LLY26a, Section 3]; the codimension is allowed to be any $r = n - k$ with $p(1-p)(r+1) \geq 1$.

8.1. **Regularized distances associated with positive contractions.** Let P be an $n \times n$ Hermitian matrix satisfying $0 \leq P \leq I$. For $\eta > 0$ and $u \in \mathbb{C}^n$, define

$$f_P(x) := \sqrt{\eta^2 + \langle P(x+u), x+u\rangle}.$$

Let $\boldsymbol{\xi} = (\xi_1, \dots, \xi_n)$ have i.i.d. Bernoulli(p) coordinates, let X be a uniform d-subset row with $p = d/n$, and put

$$\mu_P := \mathbb{E} f_P(\boldsymbol{\xi}),$$
$$D_P^2 := \eta^2 + p(1-p)\operatorname{tr} P + \langle P(u + p\mathbf{1}), u + p\mathbf{1}\rangle\,. \tag{8.1}$$

Lemma 8.1 (Convex concentration on a product cube)**.** *Let $K > 0$, let $\zeta_1, \dots, \zeta_n$ be independent complex random variables with $|\zeta_j| \leq K$ almost surely, and let $F : \mathbb{C}^n \to \mathbb{R}$ be convex and 1-Lipschitz. If m_F is a median of $F(\zeta)$, then, for every $t > 0$,*

$$\mathbb{P}\{|F(\zeta) - m_F| \geq tK\} \leq Ce^{-ct^2}, \tag{8.2}$$
$$\mathbb{P}\{|F(\zeta) - \mathbb{E}F(\zeta)| \geq tK\} \leq Ce^{-ct^2}. \tag{8.3}$$

The constants are absolute.

Proof. The median-centered estimate is [Tao12, Theorem 2.1.13]. To record the recentering used below, integrating (8.2) gives

$$|\mathbb{E}F(\zeta) - m_F| \leq \mathbb{E}|F(\zeta) - m_F| \leq K\int_0^\infty Ce^{-ct^2}\, dt \leq CK.$$

If $t \geq 2C$, then

$$\{|F(\zeta) - \mathbb{E}F(\zeta)| \geq tK\} \subset \{|F(\zeta) - m_F| \geq tK/2\};$$

for $0 < t < 2C$, the estimate follows after increasing the leading constant. This proves (8.3). □

Lemma 8.2 (Regularized distance concentration)**.** *Assume $D_P \geq 1$. Then*

$$c_0 D_P \leq \mu_P \leq D_P,$$

and, for every $t > 0$,

$$\mathbb{P}\{|f_P(\boldsymbol{\xi}) - \mu_P| \geq t\} \leq Ce^{-ct^2}, \tag{8.4}$$

$$\mathbb{P}\{|f_P(X) - \mu_P| \geq t\} \leq C\sqrt{d}\, e^{-ct^2}. \tag{8.5}$$

Proof. Write $x = \boldsymbol{\xi} - p\mathbf{1}$ and $u_0 = u + p\mathbf{1}$. Since $\mathbb{E}x = 0$ and the coordinates are independent,

$$\mathbb{E}f_P(\boldsymbol{\xi})^2 = \eta^2 + p(1-p)\operatorname{tr} P + \langle Pu_0, u_0\rangle = D_P^2.$$

We write out the positive-contraction variance calculation. Set

$$Q_P := \langle Px, x\rangle - p(1-p)\operatorname{tr} P, \qquad L_P := 2\Re\langle Px, u_0\rangle.$$

Put $\sigma^2 = p(1-p)$ and $m_4 = \mathbb{E}x_1^4$. Since x is real and $P = P^*$,

$$\langle Px, x\rangle = \sum_j P_{jj}x_j^2 + 2\sum_{j<\ell} \Re(P_{j\ell})x_j x_\ell.$$

The centered diagonal terms are orthogonal in L^2 to the off-diagonal monomials, and distinct off-diagonal monomials are likewise orthogonal. Hence

$$\begin{aligned}\operatorname{Var}(Q_P) &= (m_4 - \sigma^4)\sum_j P_{jj}^2 + 4\sigma^4 \sum_{j<\ell} (\Re P_{j\ell})^2 \\ &\leq C\sigma^2 \sum_{j,\ell} |P_{j\ell}|^2 = Cp(1-p)\operatorname{tr}(P^2).\end{aligned} \tag{8.6}$$

Here

$$m_4 = p(1-p)^4 + (1-p)p^4 = \sigma^2(1 - 3\sigma^2),$$

so $0 \leq m_4 - \sigma^4 \leq \sigma^2$, while $4\sigma^4 \leq \sigma^2$. Furthermore, self-adjointness under our linear-first convention gives

$$\langle Px, u_0\rangle = \langle x, Pu_0\rangle = \sum_j x_j \overline{(Pu_0)_j}.$$

Consequently,

$$\begin{aligned}\operatorname{Var}(L_P) &= 4\sigma^2 \sum_j \left(\Re(Pu_0)_j\right)^2 \\ &\leq 4\sigma^2 \|Pu_0\|_2^2 = 4p(1-p)\left\langle P^2u_0, u_0\right\rangle.\end{aligned} \tag{8.7}$$

Since $f_P(\boldsymbol{\xi})^2 - D_P^2 = Q_P + L_P$, it follows that

$$\operatorname{Var}(f_P(\boldsymbol{\xi})^2) \leq 2\operatorname{Var}(Q_P) + 2\operatorname{Var}(L_P)$$

$$\le C\left[p(1-p)\operatorname{tr}(P^2) + p(1-p)\left\langle P^2u_0, u_0\right\rangle\right]$$
$$\le C\left[p(1-p)\operatorname{tr} P + \langle Pu_0, u_0\rangle\right] \le CD_P^2.$$

Here we used $p(1-p) \le 1$, $\operatorname{tr}(P^2) \le \operatorname{tr} P$, and $P^2 \le P$. Thus, when $D_P \ge 1$,

$$\mathbb{E} f_P(\boldsymbol{\xi})^4 = D_P^4 + \operatorname{Var}(f_P(\boldsymbol{\xi})^2) \le CD_P^4.$$

Hölder's inequality yields

$$D_P^2 = \mathbb{E} f_P^2 \le (\mathbb{E} f_P)^{2/3} (\mathbb{E} f_P^4)^{1/3},$$

which implies $\mu_P \ge c_0 D_P$ for an absolute constant $c_0 > 0$. The upper bound is Jensen's inequality.

The map $x \mapsto f_P(x)$ is convex and 1-Lipschitz because $\left\|P^{1/2}\right\| \le 1$. Since the Bernoulli coordinates satisfy $|\xi_j| \le 1$, lemma 8.1, in its mean-centered form (8.3), gives (8.4).

Finally, the row-slice law is the law of $\boldsymbol{\xi}$ conditioned on $\sum_j \xi_j = d$. The local central-limit bound

$$\mathbb{P}\left\{\sum_j \xi_j = d\right\} \ge \frac{c}{\sqrt{d}}$$

follows directly from Stirling's formula; the explicit computation appears in the proof of [LLY26a, Corollary 3.5]. Dividing by this conditioning probability transfers (8.4) to (8.5). □

Lemma 8.3 (One-row logarithmic comparison)**.** *Let $0 \le r < n$ be an integer. Assume $\operatorname{tr} P \ge r + 1$ and put*

$$Q = p(1-p)(r+1) \ge 1.$$

Suppose $f_P(\boldsymbol{\xi}), f_P(X) \le U$ almost surely. Then

$$\mathbb{E}|\log f_P(X) - \log f_P(\boldsymbol{\xi})| \le C\sqrt{\frac{\log(ed)}{Q}} + C\sqrt{d}\log\left(\frac{U}{\eta}\right) e^{-cQ}. \tag{8.8}$$

Proof. Since $D_P^2 \ge Q$ and $D_P \ge 1$, lemma 8.2 gives $\mu_P \ge c_0\sqrt{Q}$. Let

$$\mathcal{G} = \{f_P(X) \ge \mu_P/2,\ f_P(\boldsymbol{\xi}) \ge \mu_P/2\}.$$

On $\mathcal{G}$,

$$|\log f_P(X) - \log f_P(\boldsymbol{\xi})| \le \frac{2}{\mu_P}|f_P(X) - f_P(\boldsymbol{\xi})|.$$

Integrating the tails in (8.4)–(8.5),

$$\mathbb{E}|f_P(\boldsymbol{\xi}) - \mu_P| \le C,$$

while

$$\mathbb{E}|f_P(X) - \mu_P| \le \int_0^\infty \min(1, C\sqrt{d}\, e^{-ct^2})\, dt \le C\sqrt{\log(ed)}.$$

Thus the contribution from $\mathcal{G}$ is at most $C\sqrt{\log(ed)/Q}$.

By lemma 8.2,

$$\mathbb{P}(\mathcal{G}^c) \le C\sqrt{d}\, e^{-c\mu_P^2} \le C\sqrt{d}\, e^{-cQ}.$$

Since $\eta \le f_P \le U$, the logarithmic difference on $\mathcal{G}^c$ is at most $\log(U/\eta)$. This proves (8.8). □

8.2. **The hybrid argument.** For a $k \times n$ matrix H, define

$$\Psi_\eta(H) := \frac{1}{2}\log\det(HH^* + \eta^2 I_k).$$

Let $R_1, \dots, R_k$ be independent uniform row slices. Let $\widetilde{R}_1, \dots, \widetilde{R}_k$ be i.i.d. Bernoulli(p) rows, and assume the two families are mutually independent. Let $u_1, \dots, u_k \in \mathbb{C}^n$ be deterministic.

Define a hybrid $H^{(i)}$ whose first i rows are $R_j + u_j$ and whose remaining rows are $\widetilde{R}_j + u_j$. At the ith replacement, let K_i be the matrix of the other $k-1$ rows and put

$$P_i = I - K_i^*(K_iK_i^* + \eta^2 I_{k-1})^{-1}K_i, \qquad \widehat{P}_i := \overline{P_i}. \tag{8.9}$$

For a candidate ith row with coordinate column $v \in \mathbb{C}^n$, let H_v denote the corresponding hybrid matrix. The Schur-complement identity, together with our convention that matrix rows act bilinearly, gives

$$\begin{aligned}\det(H_vH_v^* + \eta^2 I_k) &= \det(K_iK_i^* + \eta^2 I_{k-1})(\eta^2 + v^\top P_i \overline{v}) \\ &= \det(K_iK_i^* + \eta^2 I_{k-1})(\eta^2 + \left\langle \widehat{P}_i v, v\right\rangle).\end{aligned} \tag{8.10}$$

Indeed, since P_i is Hermitian,

$$\left\langle \widehat{P}_i v, v\right\rangle = v^\top \widehat{P}_i^\top \overline{v} = v^\top P_i \overline{v}.$$

The determinant involving K_i is common to the two hybrids and cancels. Since $\Psi_\eta = \frac{1}{2}\log\det$, the remaining factor gives

$$\begin{aligned}\Psi_\eta(H^{(i)}) - \Psi_\eta(H^{(i-1)}) = \log\sqrt{\eta^2 + \left\langle \widehat{P}_i(R_i + u_i), R_i + u_i\right\rangle} \\ - \log\sqrt{\eta^2 + \left\langle \widehat{P}_i(\widetilde{R}_i + u_i), \widetilde{R}_i + u_i\right\rangle}.\end{aligned} \tag{8.11}$$

The matrix P_i is a positive contraction. On $\ker K_i$, it is the identity; hence

$$\operatorname{tr} P_i \ge \dim\ker K_i \ge n - k + 1 = r + 1. \tag{8.12}$$

Entrywise conjugation preserves Hermitian positivity and the contraction property. Thus $\widehat{P}_i$ is also a positive contraction; moreover, since $\operatorname{tr} P_i$ is real,

$$\operatorname{tr}\widehat{P}_i = \overline{\operatorname{tr} P_i} = \operatorname{tr} P_i \ge r + 1.$$

Also, for either row type,

$$\sqrt{\eta^2 + \left\langle \widehat{P}_i(x + u_i), x + u_i\right\rangle} \le \sqrt{\eta^2 + (\sqrt{n} + \|u_i\|_2)^2}.$$

Conditional on all other rows, $\widehat{P}_i$ is deterministic and the two candidate rows retain their respective slice and Bernoulli laws. Therefore lemma 8.3 applies with $P = \widehat{P}_i$ and

$$U = \sqrt{\eta^2 + \left(\sqrt{n} + \max_{i \le k} \|u_i\|_2\right)^2}.$$

Telescoping (8.11), taking absolute values, conditioning, and summing gives the desired result. Notice that no independence among the increments is required.

Proposition 8.4 (Generalized regularized row replacement)**.** *Let $p = d/n$, $1 \le k < n$, and $r = n - k$. Define H_{slice} from independent row slices and H_{iid} from i.i.d. Bernoulli(p) rows, with the same deterministic shifts u_i, and assume that the two row families are mutually independent. Put*

$$Q = p(1-p)(r+1), \qquad U = \sqrt{\eta^2 + \left(\sqrt{n} + \max_{i \le k} \|u_i\|_2\right)^2}.$$

If $Q \ge 1$, then

$$\frac{1}{n}\mathbb{E}|\Psi_\eta(H_{\mathrm{slice}}) - \Psi_\eta(H_{\mathrm{iid}})| \le C\sqrt{\frac{\log(ed)}{Q}} + C\sqrt{d}\log\left(\frac{U}{\eta}\right)e^{-cQ}. \tag{8.13}$$

Proof. Apply lemma 8.3 conditionally to each increment in (8.11). There are $k \le n$ increments, so division by n leaves the same bound. □

9. Hermitization and the superlogarithmic bridge

We combine theorem 7.1 and proposition 8.4 with the smallest-singular-value inputs and the optimal sparse i.i.d. circular law. Throughout this section, fix an arbitrary deterministic integer sequence $d = d(n)$ satisfying the degree assumptions of the result under consideration, and set $p_n = d(n)/n$. Every assertion is made for this fixed sequence. The Bernoulli logarithmic-potential interface below is valid for every fixed $z \neq 0$, so no sequence-dependent exceptional set of shifts is needed.

9.1. **A logarithmic-tail lemma.**

Lemma 9.1 (Negative logarithmic uniform integrability)**.** *Let*

$$\nu_n = \frac{1}{n}\sum_{j=1}^{n} \delta_{s_{j,n}}$$

be a random probability measure on $[0,\infty)$, where $s_{1,n} \ge \cdots \ge s_{n,n} \ge 0$. Suppose that, for a deterministic probability measure ν,

(i) *ν_n converges weakly to ν in probability;*

(ii)
$$\int \log t \; d\nu_n(t) \xrightarrow{\mathbb{P}} \int \log t \; d\nu(t) \in \mathbb{R};$$

(iii)
$$\int t^2 \, d\nu_n(t) = O_{\mathbb{P}}(1).$$

Then, for every $\varepsilon > 0$,
$$\lim_{T\to\infty} \limsup_{n\to\infty} \mathbb{P}\left\{ \int_{\{\log^- t > T\}} \log^- t \, d\nu_n(t) > \varepsilon \right\} = 0. \tag{9.1}$$

Consequently, if $r_n = o(n)$, *then*
$$\frac{1}{n} \sum_{j=n-r_n+1}^{n} \log^- s_{j,n} \xrightarrow{\mathbb{P}} 0. \tag{9.2}$$

Proof. We first control positive logarithms. For $R \geq e$, put
$$c_R := \sup_{t\geq R} \frac{\log t}{t^2},$$
so $c_R \to 0$. Then
$$\int_{t>R} \log t \, d\nu_n(t) \leq c_R \int t^2 \, d\nu_n(t),$$
which tends to zero in the iterated limit $n \to \infty$, $R \to \infty$, in probability. Weak convergence applied to $\min(\log^+ t, \log R)$ therefore gives
$$\int \log^+ t \, d\nu_n(t) \xrightarrow{\mathbb{P}} \int \log^+ t \, d\nu(t). \tag{9.3}$$

The bounded-in-probability second moments also imply
$$\int t^2 \, d\nu(t) < \infty.$$
Indeed, for each $L < \infty$, weak convergence in probability gives
$$\int (t^2 \wedge L) \, d\nu_n(t) \xrightarrow{\mathbb{P}} \int (t^2 \wedge L) \, d\nu(t).$$
Choose $M < \infty$ such that
$$\limsup_{n\to\infty} \mathbb{P}\left\{ \int t^2 \, d\nu_n(t) > M \right\} < \frac{1}{2}.$$
Because the truncated integral is bounded above by the full second moment, its deterministic probability limit cannot exceed M. Hence
$$\int (t^2 \wedge L) \, d\nu(t) \leq M$$
for every L, and monotone convergence proves the claim.

Since $\log t = \log^+ t - \log^- t$, assumption (ii) and (9.3) imply
$$\int \log^- t \, d\nu_n(t) \xrightarrow{\mathbb{P}} \int \log^- t \, d\nu(t) < \infty. \tag{9.4}$$
For fixed T, the function
$$f_T(t) = \min(\log^- t, T), \qquad f_T(0) = T,$$

is bounded and continuous. Thus

$$\int f_T \, d\nu_n \xrightarrow{\mathbb{P}} \int f_T \, d\nu.$$

Subtracting from (9.4) yields

$$\int (\log^- - f_T) \, d\nu_n \xrightarrow{\mathbb{P}} \int (\log^- - f_T) \, d\nu.$$

The deterministic right-hand side tends to zero by monotone convergence. On $\{\log^- t > 2T\}$,

$$\log^- t \le 2(\log^- t - T) = 2(\log^- t - f_T(t)).$$

This proves (9.1), after renaming $2T$ as T.

Finally, for fixed T,

$$\frac{1}{n} \sum_{j=n-r_n+1}^{n} \log^- s_{j,n} \le T\frac{r_n}{n} + \int_{\{\log^- t > T\}} \log^- t \, d\nu_n(t).$$

Let $n \to \infty$ and then $T \to \infty$. □

9.2. **The Bernoulli hermitization interface.** Let B_n denote an $n \times n$ matrix with i.i.d. Bernoulli(d/n) entries. For fixed $z \in \mathbb{C}$, write

$$H_{n,z} := \frac{B_n}{\sqrt{d}} - zI_n, \qquad \nu^B_{n,z} := \frac{1}{n}\sum_{j=1}^{n} \delta_{s_j(H_{n,z})}.$$

Let ν^G_z denote the deterministic limiting singular-value law of $n^{-1/2}G_n - zI_n$, where G_n has i.i.d. standard complex Gaussian entries, and put

$$U^\circ(z) := -\int_{\mathbb{C}} \log|z-w| \; d\mu_{\mathrm{circ}}(w) = -\int_0^\infty \log t \, d\nu^G_z(t). \tag{9.5}$$

Proposition 9.2 (Bernoulli hermitization interface)**.** *Assume $d \to \infty$ and $d = o(n)$. For every fixed $z \in \mathbb{C} \setminus \{0\}$:*

(i) $\nu^B_{n,z} \Rightarrow \nu^G_z$ *in probability;*

(ii)
$$-\int_0^\infty \log t \, d\nu^B_{n,z}(t) \xrightarrow{\mathbb{P}} U^\circ(z), \tag{9.6}$$
equivalently, the full untruncated logarithmic integral converges to the finite value $\int \log t \, d\nu^G_z(t)$;

(iii)
$$\int_0^\infty t^2 \, d\nu^B_{n,z}(t) = O_{\mathbb{P}}(1). \tag{9.7}$$

Proof. Part (i) is [SSS25, Fact 10.2], applied to the deterministic sequence $p_n = d/n$ and to the Bernoulli model. The Gaussian limit in that fact is ν^G_z. By [SSS25, Fact 10.1], the measure ν^G_z is compactly supported and satisfies

$$\nu^G_z([0,t)) \le Ct, \qquad t > 0,$$

and (9.5) holds. The mass bound at zero and compact support imply that the limiting logarithmic integral is finite.

We reproduce the squeeze that gives the full, untruncated convergence in part (ii). Define

$$U_n(z) := -\frac{1}{n}\sum_{j=1}^{n} \log s_j(H_{n,z}), \tag{9.8}$$

with the extended value $+\infty$ when $H_{n,z}$ is singular. Choose a deterministic sequence $\varepsilon_n \downarrow 0$ sufficiently slowly for the estimates in Sections 8–10 of [SSS25], and write $\varepsilon = \varepsilon_n$. Set

$$m = \lfloor (1-\varepsilon)n \rfloor, \qquad \ell = \lfloor (1-\varepsilon/4)m \rfloor, \qquad a = \lceil 2\varepsilon n \rceil, \qquad b = \lfloor (1-\varepsilon)n \rfloor.$$

Let $H_{m,z}$ be the top-left $m \times m$ principal minor of $H_{n,z}$. Thus its entries retain the normalization $d^{-1/2}$ used for $H_{n,z}$. Define

$$T_n(z) := -\frac{1}{n}\sum_{j=1}^{\ell} \log s_j(H_{m,z}), \tag{9.9}$$

$$T_n^{(1)}(z) := -\frac{1}{n}\sum_{j=1}^{\ell} \log s_j(H_{n,z}), \tag{9.10}$$

$$T_n^{(2)}(z) := -\frac{1}{n}\sum_{j=a}^{b} \log s_j(H_{n,z}) - \frac{\ell - \lfloor (1-3\varepsilon)n \rfloor}{n} \log s_\ell(H_{m,z}). \tag{9.11}$$

These are the integer-rounded versions of the quantities in Sections 8–10 of [SSS25].

For every fixed $z \neq 0$, [SSS25, Lemma 8.2] gives

$$U_n(z) \leq T_n(z) + o_{\mathbb{P}}(1). \tag{9.12}$$

The first deterministic comparison in [SSS25, Fact 9.1] gives

$$T_n(z) \leq T_n^{(2)}(z). \tag{9.13}$$

For completeness, we derive the lower comparison directly from monotonicity of the ordered singular values. Put

$$x_j = -\log s_j(H_{n,z}), \qquad 1 \leq j \leq n.$$

Since the singular values are nonincreasing, $x_1 \leq \cdots \leq x_n$. Hence

$$U_n(z) = \frac{1}{n}\sum_{j=1}^{n} x_j \geq \frac{1}{\ell}\sum_{j=1}^{\ell} x_j = \frac{n}{\ell} T_n^{(1)}(z). \tag{9.14}$$

Moreover,

$$\frac{\ell}{n} = (1-\varepsilon/4)(1-\varepsilon) + O(n^{-1}).$$

Finally, [SSS25, Lemma 9.3], proved there from Facts 10.1–10.2 and Lemmas 10.3–10.5, yields

$$T_n^{(1)}(z) \xrightarrow{\mathbb{P}} U^{\circ}(z), \qquad T_n^{(2)}(z) \xrightarrow{\mathbb{P}} U^{\circ}(z) \tag{9.15}$$

when $\varepsilon_n \downarrow 0$ sufficiently slowly. Since $n/\ell \to 1$, (9.14) and the first convergence in (9.15) give

$$U_n(z) \geq U^\circ(z) - o_{\mathbb{P}}(1).$$

On the other hand, (9.12), (9.13), and the second convergence in (9.15) give

$$U_n(z) \leq U^\circ(z) + o_{\mathbb{P}}(1).$$

Therefore $U_n(z) \to U^\circ(z)$ in probability. In particular,

$$\mathbb{P}\{H_{n,z} \text{ is singular}\} \to 0.$$

Redefining $U_n(z)$ arbitrarily on that event does not change the conclusion. This proves (9.6) for every fixed $z \neq 0$.

For part (iii),

$$\begin{aligned}\int t^2 \, d\nu^B_{n,z}(t) &= \frac{1}{n} \|H_{n,z}\|^2_{\mathrm{HS}} \\ &\leq \frac{2}{nd} \sum_{i,j=1}^{n} (B_n)_{ij} + 2\,|z|^2 .\end{aligned}$$

The sum has mean nd and is $O_{\mathbb{P}}(nd)$, proving (9.7). □

Corollary 9.3 (Bernoulli bottom tail)**.** *Fix $z \in \mathbb{C} \setminus \{0\}$. For every deterministic integer sequence $r_n = o(n)$,*

$$\frac{1}{n} \sum_{j=n-r_n+1}^{n} \log^- s_j\Big(\frac{B_n}{\sqrt{d}} - zI_n\Big) = o_{\mathbb{P}}(1). \tag{9.16}$$

Proof. Apply lemma 9.1 to the three conclusions of proposition 9.2. □

9.3. **Lower-band regularized comparison of a large prefix.** Fix $\sigma \in (0,1)$, assume (2.1), fix $z \in \mathbb{C} \setminus \{0\}$, and define the unnormalized shifted matrices

$$A = M_n - z\sqrt{d}\, I_n, \qquad \widetilde{A} = B_n - z\sqrt{d}\, I_n. \tag{9.17}$$

Here B_n is independent of M_n and has i.i.d. Bernoulli(p) entries, $p = d/n$.

By proposition 2.4, for a sufficiently large absolute constant C_L,

$$\tau_n = n^{-C_L \log\log n} \tag{9.18}$$

satisfies

$$\mathbb{P}(\mathcal{L}_n(z)^c) = O(d^{-1/2}) = o(1), \qquad \mathcal{L}_n(z) = \{s_n(A) \geq \tau_n\} \cap \Big\{s_n(\widetilde{A}) \geq \tau_n\Big\}. \tag{9.19}$$

The shift restriction in proposition 2.4 holds because $|z|\sqrt{d} \leq \sqrt{d}\log\log d$ for fixed $z \neq 0$ and all sufficiently large n.

Choose

$$r = \left\lceil \frac{n}{\sqrt{d}} \right\rceil, \qquad k = n - r, \qquad \eta = \frac{\tau_n}{n}. \tag{9.20}$$

Let C_M and C_B be the first k rows of A and $\widetilde{A}$, respectively. In proposition 8.4,

$$Q_n = p(1-p)(r+1) \asymp \sqrt{d},$$

and the deterministic shifts are $u_i = -z\sqrt{d}\, e_i$. Thus

$$U \le \sqrt{\eta^2 + (\sqrt{n} + |z|\sqrt{d})^2} \le 2(1 + |z|)\sqrt{n}$$

and

$$\log \frac{U}{\eta} = O(\log n \, \log\log n) = O(d \log d). \tag{9.21}$$

Consequently,

$$\begin{aligned}\frac{1}{n}\mathbb{E}\,|\Psi_\eta(C_M) - \Psi_\eta(C_B)| &\le O\Big(d^{-1/4}\sqrt{\log d}\Big) + O\Big(d^{3/2}\log d\, e^{-c\sqrt{d}}\Big) \\ &= o(1). \end{aligned} \tag{9.22}$$

Markov's inequality gives

$$\frac{1}{n}\,|\Psi_\eta(C_M) - \Psi_\eta(C_B)| = o_{\mathbb{P}}(1). \tag{9.23}$$

9.4. **Removing the lower-band regularization.** Let $t_1(C) \ge \cdots \ge t_k(C)$ be the singular values of a $k \times n$ prefix. Cauchy interlacing gives

$$t_i(C_M) \ge s_{i+r}(A) \ge s_n(A),$$

and similarly for C_B. On $\mathcal{L}_n(z)$, every prefix singular value is at least τ_n. Hence

$$\begin{aligned} 0 \le \Psi_\eta(C) - \log \mathrm{Vol}(C) &= \frac{1}{2}\sum_{i=1}^{k} \log\left(1 + \frac{\eta^2}{t_i(C)^2}\right) \\ &\le \frac{k\eta^2}{2\tau_n^2} \le \frac{1}{2n}. \end{aligned}$$

After division by n, the error is at most $1/(2n^2)$. Combining this with (9.23) and (9.19), and extending both logarithmic-volume random variables by zero on $\mathcal{L}_n(z)^c$, gives

$$\frac{1}{n}\,|\log \mathrm{Vol}(C_M) - \log \mathrm{Vol}(C_B)| = o_{\mathbb{P}}(1). \tag{9.24}$$

9.5. **From lower-band prefix volumes to square determinants.** Let H be an invertible $n \times n$ matrix, and let C be its first $n - r$ rows. Interlacing implies

$$\sum_{j=r+1}^{n} \log s_j(H) \le \log \mathrm{Vol}(C) \le \sum_{j=1}^{n-r} \log s_j(H).$$

Therefore

$$|\log|\det H| - \log \mathrm{Vol}(C)| \le \sum_{j=1}^{r} \log^+ s_j(H) + \sum_{j=n-r+1}^{n} \log^- s_j(H). \tag{9.25}$$

For the positive logarithms, concavity gives

$$\frac{1}{n}\sum_{j=1}^{r} \log^+ s_j(H) \le \frac{r}{2n} \log\left(1 + \frac{\|H\|_{\mathrm{HS}}^2}{r}\right). \tag{9.26}$$

Since $\|A\|_{\mathrm{HS}}^2 \le 2(1+|z|^2)nd$ deterministically and $\left\|\widetilde{A}\right\|_{\mathrm{HS}}^2 = O_{\mathbb{P}}(nd)$,

$$\frac{1}{n}\sum_{j=1}^{r}\log^+ s_j(A) = O\Big(\frac{\log d}{\sqrt{d}}\Big) = o(1), \tag{9.27}$$

with the analogous $o_{\mathbb{P}}(1)$ estimate for $\widetilde{A}$.

For the negative row-slice tail, use bottom-up indexing $a_j = s_{n-j+1}(A)$. On $\mathcal{L}_n(z)$ and the event in theorem 7.1,

$$\frac{1}{n}\sum_{j=1}^{q_0}\log^- a_j \le \frac{q_0}{n}\log\frac{1}{\tau_n} = O\Big(\frac{\log n\,\log\log n}{d^2}\Big) = o(1), \tag{9.28}$$

and

$$\frac{1}{n}\sum_{j=q_0+1}^{r}\log^- a_j \le \frac{r}{n}C_{\mathrm{isv}}(z)d^{1/96} = O(d^{-47/96}) = o(1). \tag{9.29}$$

Thus

$$\frac{1}{n}\sum_{j=n-r+1}^{n}\log^- s_j(A) = o_{\mathbb{P}}(1). \tag{9.30}$$

By corollary 9.3, and because multiplication by $\sqrt{d} \ge 1$ can only decrease $\log^-$,

$$\frac{1}{n}\sum_{j=n-r+1}^{n}\log^- s_j(\widetilde{A}) = o_{\mathbb{P}}(1). \tag{9.31}$$

Applying (9.25) to (A, C_M) and $(\widetilde{A}, C_B)$, and using (9.24), yields

$$\frac{1}{n}\log|\det A| - \frac{1}{n}\log\left|\det\widetilde{A}\right| \xrightarrow{\mathbb{P}} 0. \tag{9.32}$$

On the event where a determinant vanishes, define its normalized logarithm arbitrarily; by (9.19), this changes a random variable only on an event of probability $o(1)$. Since the common factor $d^{-n/2}$ cancels,

$$\frac{1}{n}\log\left|\det\Big(\frac{M_n}{\sqrt{d}} - zI_n\Big)\right| - \frac{1}{n}\log\left|\det\Big(\frac{B_n}{\sqrt{d}} - zI_n\Big)\right| \xrightarrow{\mathbb{P}} 0 \tag{9.33}$$

for every fixed $z \ne 0$, hence for Lebesgue-almost every fixed shift.

9.6. **The lower-polylogarithmic circular law.**

Theorem 9.4 (Lower-polylogarithmic regime)**.** *Fix $\sigma \in (0,1)$. There exists an absolute constant $C_* > 0$ such that, whenever*

$$C_*\log n \le d \le (\log n)^{1+\sigma},$$

one has

$$\mu_{M_n/\sqrt{d(1-d/n)}} \xrightarrow{\mathbb{P}} \mu_{\mathrm{circ}}.$$

The constant C_* is independent of σ; the threshold from which the conclusion holds may depend on σ.

Proof. Set

$$X_n = \frac{M_n}{\sqrt{d}}, \qquad Y_n = \frac{B_n}{\sqrt{d}}.$$

Then

$$\frac{1}{n}\|X_n\|_{\mathrm{HS}}^2 = 1$$

deterministically and

$$\mathbb{E}\frac{1}{n}\|Y_n\|_{\mathrm{HS}}^2 = 1.$$

Thus the Hilbert–Schmidt condition in the replacement principle [TV10, Theorem 2.1] is satisfied. Equation (9.33) gives its logarithmic-determinant condition for every fixed $z \neq 0$, and therefore for Lebesgue-almost every fixed z. No estimate uniform in z and no union bound over shifts is required. Consequently,

$$d_{\mathrm{BL}}(\mu_{X_n}, \mu_{Y_n}) \xrightarrow{\mathbb{P}} 0. \tag{9.34}$$

The optimal sparse i.i.d. circular law [RT19, SSS25] applies because

$$p = d/n \to 0, \qquad np = d \to \infty,$$

and gives

$$d_{\mathrm{BL}}(\mu_{Y_n}, \mu_{\mathrm{circ}}) \xrightarrow{\mathbb{P}} 0.$$

The triangle inequality therefore yields

$$\mu_{M_n/\sqrt{d}} \xrightarrow{\mathbb{P}} \mu_{\mathrm{circ}}. \tag{9.35}$$

For every square matrix H, Schur triangularization gives

$$\sum_{j=1}^{n} |\lambda_j(H)|^2 \leq \|H\|_{\mathrm{HS}}^2 .$$

Consequently,

$$\int |z| \; d\mu_H(z) \leq \left(\frac{1}{n}\|H\|_{\mathrm{HS}}^2\right)^{1/2}.$$

Let $c_n = (1 - d/n)^{-1/2} \to 1$ and write $\mu_n = \mu_{M_n/\sqrt{d}}$. For any bounded 1-Lipschitz function f,

$$\begin{aligned}\left|\int f(c_n z)\; d\mu_n(z) - \int f(z)\; d\mu_n(z)\right| &\leq |c_n - 1| \int |z| \; d\mu_n(z) \\ &\leq |c_n - 1| \left(\frac{1}{n}\left\|M_n/\sqrt{d}\right\|_{\mathrm{HS}}^2\right)^{1/2} \\ &= |c_n - 1| \, .\end{aligned}$$

Thus the bounded-Lipschitz distance between the ESDs at the two normalizations tends to zero deterministically. Combining this with (9.35) proves the theorem. □

9.7. **The superlogarithmic bridge.**

Proposition 9.5 (Superlogarithmic bridge)**.** *Fix $\varepsilon > 0$. Let $d = d(n)$ be an integer sequence satisfying*

$$(\log n)^{1+\varepsilon} \le d, \qquad d = o(n).$$

Then

$$\mu_{M_n/\sqrt{d(1-d/n)}} \xrightarrow{\mathbb{P}} \mu_{\mathrm{circ}}.$$

Proof. For all sufficiently large n, one has $d \le n/2$. Let B_ε be supplied by corollary 2.5, fix $z \in \mathbb{C} \setminus \{0\}$, and define $A, \widetilde{A}$ by (9.17). Put

$$\tau_{n,\varepsilon} = n^{-B_\varepsilon}, \qquad \mathcal{L}_{n,\varepsilon}(z) = \{s_n(A) \ge \tau_{n,\varepsilon}\} \cap \left\{s_n(\widetilde{A}) \ge \tau_{n,\varepsilon}\right\}.$$

By corollary 2.5,

$$\mathbb{P}(\mathcal{L}_{n,\varepsilon}(z)^c) = O(d^{-1/2}) = o(1). \tag{9.36}$$

Choose the balanced exponent

$$a = a_\varepsilon := \frac{2+\varepsilon}{2(1+\varepsilon)}. \tag{9.37}$$

Then

$$1 - a = \frac{\varepsilon}{2(1+\varepsilon)}, \qquad a(1+\varepsilon) - 1 = \frac{\varepsilon}{2}. \tag{9.38}$$

Set

$$r = \left\lceil \frac{n}{d^a} \right\rceil, \qquad k = n - r, \qquad \eta = \frac{\tau_{n,\varepsilon}}{n}. \tag{9.39}$$

Let $C_M^{(\varepsilon)}$ and $C_B^{(\varepsilon)}$ be the first k rows of A and $\widetilde{A}$. Since $p = d/n = o(1)$,

$$Q = p(1-p)(r+1) \asymp d^{1-a} \ge c(\log n)^{\varepsilon/2} \to \infty. \tag{9.40}$$

Moreover, $U \le 2(1+|z|)\sqrt{n}$ and

$$\log \frac{U}{\eta} = O_\varepsilon(\log n).$$

Therefore proposition 8.4 gives

$$\begin{aligned} \frac{1}{n}\mathbb{E}\left|\Psi_\eta(C_M^{(\varepsilon)}) - \Psi_\eta(C_B^{(\varepsilon)})\right| &\le C\sqrt{\frac{\log(ed)}{d^{1-a}}} + C_\varepsilon \sqrt{d}\,\log n\, e^{-cd^{1-a}} \\ &= o(1). \end{aligned} \tag{9.41}$$

Indeed, $\log d = o(d^{1-a})$. Also

$$\log(\sqrt{d}\log n) = \tfrac{1}{2}\log d + \log\log n = O_\varepsilon(\log d) = o(d^{1-a}),$$

where $d \ge (\log n)^{1+\varepsilon}$ is used in the middle estimate. Thus the exponential term dominates its prefactor uniformly in the stated range.

Markov's inequality and the same interlacing calculation used in the lower band yield, after extending both log-volume random variables by zero on $\mathcal{L}_{n,\varepsilon}(z)^c$,

$$\frac{1}{n}\left|\log \mathrm{Vol}(C_M^{(\varepsilon)}) - \log \mathrm{Vol}(C_B^{(\varepsilon)})\right| = o_{\mathbb{P}}(1). \tag{9.42}$$

Here the regularization is removed on $\mathcal{L}_{n,\varepsilon}(z)$ exactly as before because $\eta = \tau_{n,\varepsilon}/n$.

It remains to pass from prefixes to square determinants. By (9.26),

$$\frac{1}{n}\sum_{j=1}^{r} \log^{+} s_j(A) = O(d^{-a}\log d) = o(1), \tag{9.43}$$

with the analogous $o_{\mathbb{P}}(1)$ estimate for $\widetilde{A}$. On $\mathcal{L}_{n,\varepsilon}(z)$, every one of the bottom r row-slice singular values may pay the polynomial smallest-singular-value cost:

$$\begin{aligned}\frac{1}{n}\sum_{j=n-r+1}^{n} \log^{-} s_j(A) &\le \frac{r}{n} B_\varepsilon \log n \\ &\le C_\varepsilon d^{-a}\log n \le C_\varepsilon (\log n)^{-\varepsilon/2} = o(1),\end{aligned} \tag{9.44}$$

where (9.38) is used in the last step. Since $r = o(n)$, corollary 9.3 gives the corresponding estimate for $B_n/\sqrt{d} - zI_n$; multiplication by $\sqrt{d} \ge 1$ can only decrease $\log^{-}$, so the same estimate holds for $\widetilde{A}$.

Applying (9.25) to both matrices and using (9.42), we obtain

$$\frac{1}{n}\log\left|\det\left(\frac{M_n}{\sqrt{d}} - zI_n\right)\right| - \frac{1}{n}\log\left|\det\left(\frac{B_n}{\sqrt{d}} - zI_n\right)\right| \xrightarrow{\mathbb{P}} 0 \tag{9.45}$$

for every fixed $z \neq 0$.

The Hilbert–Schmidt estimates, the replacement principle, and the optimal sparse i.i.d. circular law now give

$$\mu_{M_n/\sqrt{d}} \xrightarrow{\mathbb{P}} \mu_{\mathrm{circ}}$$

exactly as in the proof of theorem 9.4. Finally, $d = o(n)$ implies $(1 - d/n)^{-1/2} \to 1$, and the same Schur/Hilbert–Schmidt argument transfers the conclusion to the variance normalization. □

Remark 9.6 (Relation to the Li–Litvak–Yu range). For every fixed $\eta \in (0,1)$, [LLY26a, Theorem 1.3] already gives the circular law when

$$d \ge (\log n)^{2+\eta}, \qquad d = o(n).$$

Thus, to connect the lower band to the previously available result, proposition 9.5 is needed only on the intervening range between $(\log n)^{1+\varepsilon}$ and a threshold of the form $(\log n)^{2+\eta}$. We stated and proved the bridge for the entire superlogarithmic range because the same argument is uniform there and requires no additional input.

9.8. **Completion of the main theorem.**

Proof of theorem 1.1. Fix $\sigma = 1/2$ and split the indices into two sets:

$$d \le (\log n)^{3/2} \qquad \text{and} \qquad d > (\log n)^{3/2}.$$

On the first set, apply theorem 9.4. On the second set, apply proposition 9.5 with $\varepsilon = 1/2$. Each subsequence converges in probability to the same deterministic circular law. Therefore the original sequence does as well. □

10. Further directions

10.1. **Why the lower-band argument is logarithmic.** The superlogarithmic bridge removes any need to run the structural machinery throughout the full range $d = o(n)$ and supplies the missing connection to the Li–Litvak–Yu circular-law regime. Within the lower polylogarithmic band, where the LLTTY discretization, the codimension choice $m \asymp n/d^3$, the Gaussian-normal source range, and the conditioning-to-slice estimate are used, the numerical structural and intermediate-singular-value budgets improve with d:

$$q_0 = O(n/d^2), \qquad \frac{q_0}{n}\, d \log d = O\!\left(\frac{\log d}{d}\right) \to 0, \qquad \delta_L v_L \asymp \frac{d^{1/192}}{\log d} \gg \log d.$$

The explicit lower logarithmic restriction enters in lemma 3.2. In the present ℓ-decomposition,

$$\operatorname{diam}(\operatorname{supp} B) \le \log_2 n + O(\log d),$$

whereas that lemma requires the diameter to be $O(d)$. Below logarithmic degree, the exact occupancy functional

$$\Phi_d(B) = \mathbb{E} 2^{-\max_{a \le d} B_a}$$

can no longer be compressed uniformly into $2^{-\mathbb{E}B} e^{-cD(B)}$. A sublogarithmic proof would need to retain $\Phi_d(B)$ or establish a more refined entropy–occupancy inequality that separates bins of mass above and below $1/d$.

The current proof also does not reach every fixed constant in $d = C \log n$. The support-diameter inequality and the imported restricted-norm and smallest-singular-value estimates require a sufficiently large absolute lower constant. Reducing that constant is a quantitative problem distinct from passing from the logarithmic band to larger degrees, which is handled by proposition 9.5.

A separate question concerns the sharp scale of the unshifted smallest singular value. Li, Litvak, and Yu proved $s_n(M_n) = O_{\mathbb{P}}(\sqrt{d}/n)$ in the dense regime $d = pn$ for each fixed $p \in (0, 1/2]$, and conjectured the same upper scale whenever $\min(d, n-d) \ge (1+\varepsilon)\log n$ [LLY26b, Theorem 1.1, Remark 1.4, and Conjecture 1.5]. The present argument instead uses lower bounds for shifted matrices and controls a block of intermediate singular values; it neither resolves nor requires that sharp-scale conjecture.

10.2. **Defective columns below the invertibility threshold.** A fixed column of M_n is zero with probability

$$\left(1 - \frac{d}{n}\right)^n = \exp\!\left(-d + O\!\left(\frac{d^2}{n}\right)\right) = e^{-(1+o(1))d},$$

where the last equality uses $d = o(n)$. Thus below the logarithmic scale the matrix has zero columns with substantial probability, and $s_n(M_n) = 0$ at $z = 0$. This does not by itself rule out the circular law when $d \to \infty$, because the proportion of zero columns is $e^{-(1+o(1))d} \to 0$ and hermitization is an almost-everywhere statement in the shift. It does show that any sublogarithmic proof must tolerate an exceptional coordinate set rather than demand global invertibility or expansion.

The structural theorem already allows an exceptional set of size

$$nd^{-1/192+o(1)},$$

much larger than the expected number

$$n \exp\left(-d + O\left(\frac{d^2}{n}\right)\right)$$

of zero columns when $d \to \infty$. This suggests a defect-tolerant version of the present argument: first remove low-degree columns, then prove many-level geometry and random-normal anti-clustering on the complement. The operator-norm scale also changes from $O(\sqrt{d})$ to $O(\sqrt{d} + \log n)$, so the net radii would have to be rebalanced.

10.3. **Conjectural optimal range.**

Conjecture. The independent row-slice model obeys the circular law whenever

$$d \to \infty, \qquad d = o(n).$$

This is consistent with the i.i.d. Bernoulli and doubly regular models. For fixed d, a positive limiting proportion of zero columns prevents convergence to the circular law. Theorem 1.1 reaches every sparse degree above a sufficiently large logarithmic constant. The remaining asymptotic gap is therefore the sublogarithmic regime, together with the quantitative question of how small the constant in $d = C \log n$ can be. The present paper identifies two ingredients likely needed below logarithmic degree: an occupancy-sensitive replacement for the support-diameter max-weight lemma and a defect-tolerant structural theorem.

Appendix A. Source extraction for the LLTTY deterministic consequences

This appendix proves proposition 2.2 as an author-derived result. Page and equation locators refer to [LLTTY19], arXiv:1801.05575v4. The table below separates exact source statements, immediate corollaries, and conclusions reconstructed from source proofs. No item is presented as a verbatim theorem unless it is one.

Item	Source locator	Status	Deduction recorded here
(2.8) and class count	pp. 34–35, Lemma 5.1, (14)–(16); p. 37, Lemmas 5.4–5.5	direct restatement after reindexing	Restriction to $\mathcal{A}_k(\mathcal{X})$ cannot increase the number of nonempty equivalence classes.
$S_L = \bigcup K_u(L) \cup P_v(L)$	p. 38, Theorem 5.6	verbatim after notation change	Codimension enters only through the previously defined gradual class.
(2.9)	p. 55, Lemma 7.2	stated "in particular" consequence	Reindex the four source families by the manuscript's part index q.
(2.10)	p. 61, start of Case 2 in the proof of Lemma 7.7	proof-level consequence	Each of four geometrically decreasing families contributes $O(d^{-\kappa}n)$ below the cutoff.
(K1) and (2.11)	pp. 60–61, proof of Lemma 7.7, source (32)–(33), (38), Case 1	author-derived packaging	Multiply the three source families and use $n/s_q \le d^{\kappa}$.
(K2) and (2.12)	p. 61, Case 2 in the proof of Lemma 7.7	intermediate proof conclusion	Define the first lower and last upper occupied weight orders and divide the total small-part mass by the per-order bound.
Low-order conclusion in part (v)	p. 62, Lemma 7.8	direct source conclusion	A repeated rounded value produces a weight set of mass at least $\delta/36$.
High-order conclusion and (2.15)	pp. 62–63, Lemma 7.9 and its proof	intermediate proof conclusion	The height lower bound and $\widetilde{w}_q \ge h_q/d$ yield the separated upper order.

Throughout the appendix, the source's gradual class S is our codimension-dependent class S_L, and its families K_u, P_u are denoted by $K_u(L), P_u(L)$. The codimension enters only through the definition of the steep classes and hence through S_L; the arguments in source Sections 5–7 are otherwise unchanged.

For a fixed canonical k-approximation, reindex all nonempty regular and spread ℓ-parts as $L^{(q)}$, and write

$$s_q = \left|L^{(q)}\right|, \qquad h_q = h(L^{(q)}).$$

The source's truncated weight $\widetilde{w}_q$ is exactly (2.4); its w-set W_b is the union of the parts with $2^b \le \widetilde{w}_q < 2^{b+1}$. Thus the random dyadic order B used here satisfies

$$\mathbb{P}\{B = b\} = \frac{|W_b|}{n}.$$

No rescaling or rebinning of the source weights is made.

A.1. **Equivalence classes and the gradual decomposition.** Source Lemma 5.1 states that, when $d \le n^{1/3}$ and $1 \le k \le \sqrt{n}/d^{3/2}$, the normalized canonical k-approximations form at most e^n equivalence classes under equality of the coordinate multiset. This is precisely the relation $\sim_k$ on $\mathcal{A}_k(L)$. Restricting an equivalence relation to a subset cannot increase the number of nonempty classes, which gives the corresponding assertion for $\mathcal{A}_k(\mathcal{X})$.

Source Lemma 5.4 shows that permutation of a canonical approximation preserves, order by order, the type, height, and cardinality of every nonempty ℓ-part. Hence these data are well defined on each equivalence class. Source Lemma 5.5 then separates spread and regular parts by their orders and gives, in its notation,

$$|\mathcal{C}| \le n! \prod_{j \ge 0} \frac{(h_j^S)^{c_j^S} (h_j^R)^{c_j^R}}{c_j^S! c_j^R!}.$$

After reindexing every nonempty spread or regular part by q, this is exactly

$$|\mathcal{C}| \le \frac{n!}{\prod_q s_q!} \prod_q h_q^{s_q},$$

which is (2.8). Finally, source Theorem 5.6 states, for every integer $v \ge 5$,

$$S_L = \bigcup_{u=4}^{v} K_u(L) \cup P_v(L).$$

This proves parts (i) and (ii) of proposition 2.2.

A.2. **Factorial compression inside one weight order.** Fix a dyadic order b, put

$$I(b) = \left\{ q : 2^b \le \widetilde{w}_q < 2^{b+1} \right\}, \qquad t_b = \sum_{q \in I(b)} s_q = |W_b|.$$

Source Lemma 7.2 shows that the part sizes in a fixed w-set, arranged in nonincreasing order, are controlled by a geometric sequence. Its stated "in particular" conclusion is

$$\prod_{q \in I(b)} s_q! \ge t_b! e^{-C t_b},$$

which is (2.9).

We also need the small-part consequence used in the second case of the proof of source Lemma 7.7. Let

$$T = d^{-\kappa} n, \qquad J(b) = \{ q \in I(b) : s_q < T \}.$$

The proof of source Lemma 7.2 splits the parts in one w-set into four families according to regular/spread type and large/small status. Within each family, the sizes decrease geometrically with the order at which the parts were

created. Consequently, the sum of the members of that family below the cutoff T is $O(T)$. Summing the four families gives

$$\left| \bigcup_{q \in J(b)} L^{(q)} \right| \le CT = Cd^{-\kappa} n.$$

This is also the estimate written explicitly at the start of Case 2 in the proof of source Lemma 7.7. It proves part (iii) of proposition 2.2.

A.3. **The two K_u alternatives.** Let y be the d^u-approximation of a vector in $K_u(L)$. By the source definition of K_u, the union of its spread ℓ-parts has size at least $c_K n_3$. Set

$$\kappa = \frac{c_K a_3}{16} < \frac{1}{3}, \qquad \mathscr{I} = \{q : s_q \ge d^{-\kappa} n\}.$$

Either

$$\sum_{q \in \mathscr{I}} s_q \ge n - \frac{c_K n_3}{4}$$

or its complement has mass at least $c_K n_3/4$. These are alternatives (K1) and (K2).

Assume (K1). Write $\mathscr{I}_1$ for the large spread parts, $\mathscr{I}_2$ for the large regular parts, and $\mathscr{I}_3 = (\mathscr{I}_1 \cup \mathscr{I}_2)^c$. Since the total spread mass is at least $c_K n_3$,

$$\sum_{q \in \mathscr{I}_1} s_q \ge \frac{3c_K n_3}{4}, \qquad \sum_{q \in \mathscr{I}_3} s_q \le \frac{c_K n_3}{4}.$$

For these three families, the definition of the truncated weights and the source lower bounds corresponding to its equations (32)–(33) give

$$\begin{aligned}
\widetilde{w}_q^{-s_q} &\le d^{-s_q/2} \left(\frac{n}{s_q}\right)^{s_q/2} h_q^{-s_q}, && q \in \mathscr{I}_1, \\
\widetilde{w}_q^{-s_q} &= \left(\frac{n}{s_q}\right)^{s_q} h_q^{-s_q}, && q \in \mathscr{I}_2, \\
\widetilde{w}_q^{-s_q} &\le d^{s_q} h_q^{-s_q}, && q \in \mathscr{I}_3.
\end{aligned}$$

Because $n/s_q \le d^\kappa$ on $\mathscr{I}_1 \cup \mathscr{I}_2$, the same three-family calculation as in Case 1 of the proof of source Lemma 7.7 yields

$$\begin{aligned}
\prod_q h_q^{s_q} \prod_q \widetilde{w}_q^{-s_q} &\le C^n d^{-3c_K n_3/8} d^{\kappa n} d^{c_K n_3/4} \\
&\le C^n d^{-c_K n_3/16} \le C^n d^{-c_1 n}
\end{aligned}$$

for an absolute $c_1 > 0$, after a harmless adjustment for $n_3 = \lfloor a_3 n \rfloor$. This is (2.11).

Assume now (K2). For each b, let

$$J(b) = \mathscr{I}^c \cap I(b).$$

The preceding subsection gives

$$\left| \bigcup_{q \in J(b)} L^{(q)} \right| \leq C' d^{-\kappa} n.$$

Following Case 2 in the proof of source Lemma 7.7, define b_1 to be the first order for which the cumulative lower w-sets have mass at least $c_K n_3/16$, and b_2 to be the last order for which the cumulative upper w-sets have the same mass. Then

$$\mathbb{P}\{B \leq b_1\} \geq \frac{c_K n_3}{16n}, \qquad \mathbb{P}\{B \geq b_2\} \geq \frac{c_K n_3}{16n}.$$

The small parts between these two cutoffs have total mass at least $c_K n_3/8$. Since each intervening order contributes at most $C' d^{-\kappa} n$,

$$b_2 - b_1 + 1 \geq \frac{c_K n_3}{8C' d^{-\kappa} n} \geq c d^{\kappa}.$$

After reducing fixed constants and taking n large, this gives

$$\mathbb{P}\{B \leq b_1\} \geq c_0, \qquad \mathbb{P}\{B \geq b_2\} \geq c_0, \qquad b_2 - b_1 \geq c_1 d^{\kappa}.$$

This proves part (iv) of proposition 2.2.

A.4. **The clustered P_v alternative.** Let $x \in P_v(L)$ have at least δn coordinates in a disk of radius $\rho \leq d^{-v}$, and let y be its d^v-approximation. Source Lemma 7.8 shows that one value of y is repeated on at least $\delta n/9$ coordinates and therefore produces a w-set of mass at least $\delta n/36$ and order at most

$$b_1 = \log_2 \left(\frac{72\sqrt{d}}{\delta} \right),$$

with the integer rounding used in the source.

By the source definition of P_v, at least $c_P n_3$ coordinates belong to parts whose heights are at least

$$c_P a_3 \, 2^{c_P a_3 (v-4)}.$$

Since every truncated weight satisfies $\widetilde{w}_q \geq h_q/d$, these coordinates lie in w-sets above the high order used in the proof of source Lemma 7.9. The separation hypothesis (2.14) is exactly the hypothesis under which that proof establishes the intermediate conclusion

$$b_2 - b_1 \geq \frac{c_P a_3}{2}(v-4).$$

Thus

$$\mathbb{P}\{B \leq b_1\} \geq \frac{\delta}{36}, \qquad \mathbb{P}\{B \geq b_2\} \geq \frac{c_P n_3}{n} \geq c_0$$

for all sufficiently large n, after reducing the common absolute constant c_0 used above. The size condition (2.13) and the displayed separation condition are precisely the remaining assumptions of source Lemma 7.9. This proves part (v) of proposition 2.2 and completes the source extraction.

Appendix B. Dependency ledger and source interfaces

For clarity, we list the imported results and the new arguments proved in this paper. This appendix is not logically necessary, but it makes the proof interfaces auditable.

B.1. **Imported statements.**

Li–Litvak–Yu [LLY26a]. All theorem and remark numbers cited from this preprint refer to arXiv:2604.10446v1 (April 2026). The source mappings used here are: Theorem 1.3 for the circular law above $\log^{2+\varepsilon} n$; Theorem 1.7 and Remark 1.9 for the row-slice smallest-singular-value estimate and its refined threshold; Theorem 3.1 and Remark 3.2 for the Bernoulli counterpart; and Theorem 4.1 for the rectangular restricted operator norm. The deduction of a single polynomial threshold above $(\log n)^{1+\varepsilon}$ is not quoted as a source theorem: it is carried out in corollary 2.5 from the formula in source Remark 1.9. Section 3 and Corollary 3.5 of the source are methodological antecedents for the replacement and conditioning steps, respectively; the arbitrary-codimension regularized comparison used here is proved in proposition 8.4. We do not import a codimension-L right-kernel theorem from this source.

LLTTY19 [LLTTY19]. Source checks use arXiv:1801.05575v4. Section 4 supplies the codimension-dependent steep classes $T_i(L)$, their constant-shifted versions, and the exact gradual class S_L; the particular locators used later are Section 4.1, equation (8), for the shifted very-steep inclusion, Lemma 4.5 for the shifted nets, and Lemma 4.11 for the deterministic almost-constant reduction. Sections 5–7 supply the canonical k-approximation, the ℓ-decomposition, source nets, and equivalence classes. The author-derived consequences in proposition 2.2 are extracted from Lemmas 5.1, 5.4, 5.5, 7.2, 7.7–7.9 and Theorem 5.6. A lemma-by-lemma derivation of the factorial, geometric, and clustered-P_v conclusions, including the intermediate conclusions taken from the proofs of source Lemmas 7.7–7.9, is given in appendix A. The row-local small-ball input is Lemma 6.4, and the tensorization form is Lemma 2.7. The clustered-P_v hypotheses are reproduced in (2.13)–(2.14).

LLTTY21 [LLTTY21]. Source checks use arXiv:1801.05576v2. Proposition 6.1 isolates a deterministic fixed-subspace consequence of Theorem 4.2, Propositions 4.4 and 4.7, and Lemmas 4.8–4.10. Its proof gives a separate verification of the dimension requirements, the source linear-combination count p_{src}, the cutoff $\ell_0 \le 4n/L$, and the radius, cluster-size, and steepness comparisons. The deterministic tie-break is the one explicitly suggested immediately before source Lemma 4.8. No random-regular-digraph probability estimate is imported.

Rudelson–Tikhomirov [RT19]. The optimal sparse i.i.d. circular law under the minimal divergence condition is used only as a comparator result. No intermediate estimate from that paper is imported into the row-slice structural argument.

SSS25 [SSS25]. Source checks use arXiv:2310.17600v2. Proposition 9.2 reproduces the complete logarithmic-potential squeeze needed here. Fact 10.2 supplies bounded-continuous convergence to the deterministic shifted Gaussian singular-value law ν_z^G, and Fact 10.1 identifies $U^\circ(z) = -\int \log t \, d\nu_z^G(t)$ and proves finiteness. Lemma 8.2, Fact 9.1, and Lemma 9.3 give the explicit upper and lower bounds for the full, untruncated logarithmic potential. The resulting statement holds for every fixed $z \neq 0$, rather than only on an unspecified full-measure set. The Hilbert–Schmidt bound in proposition 9.2 is elementary.

Tao's concentration theorem [Tao12]. The convex concentration statement used for regularized Bernoulli distances is Theorem 2.1.13. It gives Gaussian tails for a convex 1-Lipschitz function of bounded independent complex coordinates, centered at a median. The median-to-mean calculation is recorded in lemma 8.1.

Tao–Vu [TV10]. The replacement principle used in the lower-band and superlogarithmic arguments is Theorem 2.1. Its Hilbert–Schmidt and logarithmic-determinant hypotheses are verified explicitly in theorem 9.4 and proposition 9.5.

B.2. **New statements proved here.** The following components are specific to the independent row-slice lower-band argument and its superlogarithmic completion.

(1) The max-weight inequality, lemma 3.2.
(2) The averaged multinomial row-profile estimate and its conditioning back to a row slice, proposition 3.3 and (3.13).
(3) The entropy–energy cancellation $H(B) \le C + \log(1 + D(B))$ and the completion of the K_u and clustered-P_v class sums uniformly for $C_* \log n \le d \le (\log n)^{1+\sigma}$.
(4) The rectangular LLTTY compatibility theorem, theorem 4.4: direct independent-row exclusion of $T_0^{\mathcal{K}}(L)$, $T_1^{\mathcal{K}}(L)$, $T_2^{\mathcal{K}}(L)$, and the nonzero vectors in $\mathcal{B}(10/d^3) \setminus T_3^{\mathcal{K}}(L)$, with literal identification of the remaining nonzero kernel vectors with $\mathbb{C}^\times S_L$.
(5) The rectangular row-slice structural theorem, theorem 5.3, uniformly for codimensions $L = m$ and $L = m + 1$ throughout the lower band.
(6) The exact-slice double-counting lemma, lemma 6.2.
(7) The lower-band intermediate-singular-value bound, theorem 7.1.
(8) The generalized regularized row-replacement estimate, proposition 8.4, valid for arbitrary codimension with $p(1-p)(r+1) \ge 1$.
(9) The abstract negative-log tail lemma, lemma 9.1, and the lower-band prefix-to-square synthesis proving theorem 9.4.
(10) The superlogarithmic bridge, proposition 9.5, which combines the uniform polynomial smallest-singular-value scale with the codimension

$r = \lceil n/d^{a_\varepsilon} \rceil$ and connects the lower band to the Li–Litvak–Yu circular-law range. The proof remains valid throughout the full superlogarithmic range.

B.3. **Measurability and probability bookkeeping.** All row-set statements are uniform for each deterministic fixed set of the indicated size. Measurable enlargements are produced by the fixed standard-basis extension rule in Section 6, and greedy Gaussian clusters use the fixed total-order tie-break. For a fresh slice row, independence is stated relative to the sigma-algebra generated by the exposed rows and all auxiliary Gaussian randomness.

The structural theorem holds for each fixed exposed row set with failure $O(1/n)$, not simultaneously over all such sets. In the leave-one-row-out argument, no union bound over n indices is taken. Instead, the per-index structural failure contributes only $O(1)$ to the expected number of bad indices. Markov's inequality is then applied at the much larger threshold n/d^2.

All convergence statements in both hermitization regimes are in probability. Random errors arise only through expectation bounds followed by Markov's inequality, deterministic estimates on events of probability $1-o(1)$, or imported Bernoulli convergence in probability. The proof is carried out separately for each fixed nonzero shift z; there is no union bound over uncountably many shifts. Excluding the single point $z = 0$ is immaterial for the almost-everywhere hypothesis in the replacement principle.

For the lower band, the prefix codimension is $r = \left\lceil n/\sqrt{d} \right\rceil$, and only $q_0 = O(n/d^2)$ bottom singular values pay the full smallest-singular-value cost. In the superlogarithmic bridge, the codimension is $r = \lceil n/d^{a_\varepsilon} \rceil$, and all r bottom singular values may pay the uniform polynomial cost.

B.4. **Normalization.** All determinant comparisons are performed at the $\sqrt{d}$ normalization. The variance normalization is recovered only after the circular law has been proved:

$$\frac{M_n}{\sqrt{d(1-d/n)}} = (1-d/n)^{-1/2}\frac{M_n}{\sqrt{d}},$$

and the deterministic scalar tends to one because $d = o(n)$. This avoids invoking any unproved uniformity of the Bernoulli logarithmic-potential theorem under varying shifts.

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